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Abstract

We examine concerns that pricing algorithms, employing reinforcement learning, used by competitors would autonomously and systematically learn to collude. Findings of supracompetitive prices with Q-learning have recently raised that alarm. However, a detailed analysis of the inner workings of this algorithm type reveals that it often does not satisfy conditions for what constitutes 'autonomous' algorithmic collusion that would be a cartel risk in practice. We find that Q-learning can only learn collusive equilibria on timescales irrelevant to the firm's objective. Competitors are committed to use the same Q-learning algorithm, which start at the same moment, with the same hyperparameters and action spaces, while it is outperformed by the first alternative pricing rule. This level of synchronization suggests the need for an explicit cartel agreement. Our analysis gives criteria for practically relevant, explicitly and tacitly colluding pricing algorithms that would constitute a threat to competition. Whether autonomous algorithmic collusion is a potential threat to competition remains to be seen. There is not yet reason for competition agencies to be overly suspicious of pricing algorithms, other than of 'collusion by algorithm', in which pricing software is used to implement cartel agreements or is coded with collusive intent.

JEL-codes: C63, L13, L44, K21. Keywords: collusion, Q-learning, algorithm, pricing

1 Introduction

There is widespread concern among competition specialists and authorities around the globe that the increasing use by businesses of data-driven algorithms to determine operational decisions such as prices and production volumes present increased risks of collusion among competitors. A main worry is that such algorithms learn to collude 'autonomously', that is, without

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intent or the need for explicit coordination or communication (Ezrachi and Stucke, 2016; Mehra, 2016; OECD, 2017; Ezrachi and Stucke, 2020; Mehra, 2021). That such a spontaneous 'meeting of the artificially intelligent minds' would escape the existing cartel prohibition and enforcement has become inspiration for various proposals to change competition laws and antitrust enforcement priorities (Harrington Jr., 2018; Gal, 2019; Beneke and Mackenrodt, 2020; Bernhardt and Dewenter, 2020; Coglianese and Lai, 2022; Mazundar, 2022; Harrington Jr., 2022; Gal, 2023). Algorithmic collusion has become a call to arms for antitrust authorities that does not go unheeded (FTC and DoJ, 2017; Assad et al., 2021; Competition and Authority, 2021). Cases of raised prices are reported on on-line platforms where algorithmic sellers bid against each other, such as Bol.com (Wieting and Sapi, 2021) and Amazon (Brown and MacKay, 2021), as well as in retail gasoline markets (Byrne and De Roos, 2019; Assad et al., 2024) and hotel room pricing (OECD, 2023). In several high-profile cases, the U.S. Department of Justice submitted that the automation of anticompetitive schemes with pricing algorithms is per se illegal under the antitrust laws.[1] Related legislative proposals lower the burden of proof by defining the sharing of non-public information with an algorithm as a price-fixing agreement.[2]

A rapidly expanding literature has taken up the challenge of finding pricing algorithms that learn to collude by themselves, without explicit intent or communication. It was precipitated by the influential Calvano et al. (2020), which concludes autonomous collusion by Q-learning from a surprisingly large number of supra-competitive limit prices when two firms in Bertrand competition with differentiated goods let this algorithm make their pricing decisions. Subsequent work has reproduced this phenomenon of Q-learning algorithms finding high prices. Collusive properties of similar algorithms have been reported in various other settings, including sequential-move games (Klein, 2021), sellers on a platform (Sánchez-Cartas and Katsamakas, 2022), settings with more advanced reinforcement learning methods (Hettich, 2021; Kastius and Schlosser, 2021; Wang, 2022; Deng, Schiffer and Bichler, 2024), play against simple pricing rules (Wang, Huang and Singh, 2022), dealer markets with multiple market makers (Han, 2022; Xiong and Cont, 2021), and continuous-time models (Cartea et al., 2022, 2023). Dolgopolov (2024) analyzes the dynamics of Q-learning in the memoryless case. The general impression is that prices above the competitive level are learned because the algorithms learn to play collusive strategies by themselves. Autonomous algorithmic collusion would therefore be more than a remote theoretical possibility, and should ring an alarm bell with competition authorities.

Meanwhile, companies are increasingly making practical use of pricing algorithms, which have the potential to enhance efficiency, both on the demand- and the supply-side of their businesses,

[1] In the *Las Vegas Hotels* case, a consumer damages claim that was denied also on appeal, the U.S. Department of Justice criticized a district court for dismissing the case "based on an incorrect view of the law" (U.S. Department of Justice, 2024*a*); in the *RealPage* case, the DoJ submitted a statement of interest that "competitors' joint use of common algorithms can remove independent decision making", and subsequently prosecuted the software company for decreasing competition among landlords in apartment pricing (U.S. Department of Justice, 2023, 2024*b*); and in the *Multiplan* litigation, the DoJ clarified that the Sherman Act can cover concerted action by competitors on any "formula underlying price policies", including using a common pricing algorithm (U.S. Department of Justice, 2025).

[2] See 'The Preventing Algorithmic Collusion Act', Senate Bill 232 (S. 232), that was reintroduced in the Senate on January 23, 2025. There are also various U.S. state legislative initiatives. The regulatory responses in Europe have been more reserved and studious. *RealPage* was settled in November 2025 with requirements that the software no longer uses nonpublic data (RealPage, 2025).

for example, through reduced search efforts and transaction costs, personalized sales, lower inventory and labor expenses (MacKay, Svartbäck and Ekholm, 2023; Garcia, Tolvanen and Wagner, 2024; Spann et al., 2024), and competitive dynamic pricing (Aksoy-Pierson, Allon and Federgruen, 2013; Liu and Zhang, 2013; Levin, McGill and Nediak, 2009). In fact, smart automation holds pro-competitive promises of disrupting existing markets and creating new ones (OECD, 2023). It can make business stealing easier and faster, and thereby collusion harder to sustain. If the simulation results in stylized models lack sufficient external validity to conclude that there are clear and present dangers, lawmakers and enforcement agencies may be overly concerned with suspicions of autonomous algorithmic collusion. This can easily lead to over-enforcement and create perceived liability risks of alleged collusion, which may also deter benign and pro-competitive use of business software (Bertini and Koenigsberg, 2021).[3] Better understanding what threat algorithmic collusion may pose to competition, therefore, is an urgent and important matter. In particular, it seems relevant to be able to clearly distinguish pricing algorithms that may come to collude by themselves, without being set up or designed to, from those that facilitate intentional collusion.

In this paper, we examine whether the concern that algorithms systematically and autonomously collude follows from the phenomena observed in the literature. We focus on the seminal Calvano et al. (2020), but our analysis is applicable to a large part of the subsequent literature on pricing by reinforcement learning algorithms. Calvano et al. (2020) and the follow-up literature's conclusion that Q-learning colludes is based on their finding that the learned equilibrium strategies would involve a reward-punishment scheme that incentivizes firms to consistently price above the competitive level. Careful examination reveals, however, that this is mostly not the case. While Q-learning algorithms playing against each other eventually converge to stable limit prices, collusive equilibria do not exist on timescales relevant for the objective of the firms.[4] Also, the inference that the supra-competitive prices are sustained by reward-punishment schemes needs qualification, as a substantial fraction of the limit strategy profiles leading to supra-competitive prices do not satisfy the definition of a reward-punishment scheme, while there exist reward-punishment strategy profiles that do not sustain supra-competitive prices. In addition, if the Q-learning algorithms converge to collusive equilibria, they do so intrinsically slowly, without hope of possibilities to speed them up by hyperparameter tuning. They generate negligible extra profits for the firms committing to their use. Moreover, Q-learning is easily outperformed by reasonable alternative pricing rules available. We conclude that the current examples of supra-competitive prices do not give sufficient reason for practical concern that the use of reinforcement learning pricing algorithms would spontaneously lead to collusion unaided, without conscious coordination. The examples assume a level of initial synchronization of the software

[3]RealPage maintains after its settlement with the DoJ that its software and the value it provides in bringing together supply and demand for housing efficiently was grossly misunderstood by the government (RealPage, 2025).

[4]Throughout the article, we use the term 'collusive equilibrium' to refer to a strategy profile that is an equilibrium, in the sense that both players use a strategy that is a best response to the opponent's strategy, and contains a reward-punishment scheme. We distinguish between limit strategy profiles to refer to the strategy profiles learned in the long run and limit strategy equilibria to refer to the subset of the limit strategy profiles that are an equilibrium. By limit strategies, we refer to the strategies learned by a single player.

and a commitment to their use that suggests the need for an explicit agreement. [5]

Other papers also raise questions about aspects of algorithmic collusion. Kühn and Tadelis (2018) stress the difficulty of achieving price coordination, which they argue algorithms cannot overcome. Schwalbe (2019) identify the simplicity of the algorithm as the source of seeming collusion. Abada and Lambin (2023) examine the one-period price cuts used by Calvano et al. (2020) to infer reward-punishment schemes and suggest insufficient exploration as one of the drivers of what the authors call 'seemingly collusive outcomes'. Abada, Lambin and Tchakarov (2024) report that algorithmic sophistication induces defection to competitive prices rather than more robust price elevation. Epivent and Lambin (2024) find pricing patterns in similar algorithmic settings that suggest failure to compete rather than collusion. Eschenbaum, Mellgren and Zahn (2022) point out the lack of offline training and find that collusion breaks down when collusive reinforcement learning policies are extrapolated from a training environment to the market. Asker, Fershtman and Pakes (2021, 2022, 2024) highlight how obtaining supra-competitive limit prices strongly depends on the learning protocol of the algorithms. Cartea, Chang and Penalva (2022) find that the tick size has a significant impact on whether collusive outcomes are observed in electronic markets. Lambin (2024) examines the role that simultaneous exploration of the algorithms plays in learning supra-competitive prices, and thus calls into question the interpretation that reward-punishment schemes are causal for the supra-competitive prices. Veljanovski (2022) interprets the theoretical and empirical evidence for algorithmic collusion as being 'highly exaggerated' (p. 620). Deng (2023) reviews the literature on algorithmic collusion to conclude that it leaves many questions unanswered. Harrington Jr (2025*a*) warns against misdirected legislation.

The remainder of this paper is organized as follows. In the next section, we discuss what would be required to conclude that a reinforcement learning algorithm forms a practical threat of collusion to competitive markets, based on criteria used in the literature. To subsequently review the autonomous collusion concerns on these criteria, in Section 3 we provide a self-contained summary of Q-learning and the pricing model with just two feasible prices, which allows a full view of its inner workings. Five fundamental limitations of the literature on autonomous algorithmic collusion are then set out in detail in Section 4. In Section 5, we reflect on the limitations of the literature and discuss avenues for further progress. Section 6 concludes on the balance between risks and opportunities of algorithmic pricing with some policy recommendations.

2 Algorithmic collusion as a practical threat to competition

It is not obvious what classifies as algorithmic collusion, as currently there is no widely agreed upon definition. Just convergence to supra-competitive price levels does not seem to suffice, as it can be a feature of, for example, truncated or incomplete learning. Neither is it clear why the presence of reward-punishment schemes is necessary, as these simple algorithms do not make full infinite horizon present-discounted value trade-offs, but rather grope to improve their

[5]The role of commitment to the use of algorithms in the context of algorithmic collusion is analyzed, for example, by Dorner (2021) and Arunachaleswaran et al. (2024).

performance metric. Instead, good performance against deviations to alternative algorithms seems an attractive feature. Definition proposals based on increasingly shared concepts in the literature are made in den Boer and Meylahn (2024), Abada et al. (2024), and Bichler, Durmann and Oberlechner (2025), and are under discussion. However, several reasonable criteria that practically relevant colluding pricing algorithms should satisfy present themselves.

A specific set of three conditions for algorithmic collusion emerges from the literature on automated pricing by reinforcement learning, Calvano et al. (2020) and other early works on algorithmic collusion. Firstly, collusion should be learned on timescales relevant for the firm's decision-making and not only in the distant future. Secondly, the emergence of supra-competitive prices should not be by mistake or due to incomplete learning, but result from learning (close to) equilibrium strategy profiles. Thirdly, a requirement for learned strategies to be 'collusive' is that the learned strategies contain a reward-punishment scheme, which is based on cartel stability theory in which such schemes sustain classic collusion (Harrington Jr., 2018). In addition, it is reasonable to require that collusive algorithms perform well against reasonable alternatives that may be used by competitors, and finally that the learning of supra-competitive prices should not require coordination on behalf of the firms that itself constitutes collusion.

Therefore, for the use of particular classes of pricing algorithms to be a practical concern for competition authorities of possible collusion, they should generate the high expected present-discounted value for each user, learn sufficiently fast and completely, and be stable against deviation, both within the algorithms used and across to other pricing algorithms. In addition, models should have good responses to asymmetries in pricing rules, business environments, and hyperparameters, as well as to relevant market uncertainties or trends, such as demand and cost, which can be subject to shocks, or growth. That is, the algorithm should have a reasonable attractiveness for a firm to continue to outsource its pricing decisions to, given the pricing decisions of its rivals. Models that fall short on these characteristics lack the external validity needed to claim relevance for collusion concerns about real-world markets.

For all practical purposes, it is useful to distinguish three forms in which automated pricing software can play a collusive role in strategic interactions between competitors: autonomous, tacit, and explicit. We reserve the term 'algorithmic collusion' for autonomous collusion by algorithms that were implemented without malign intent or instruction to collude. This is opposed to 'collusion by algorithm', in which pricing software is employed in the service of reaching a cartel agreement. We distinguish two types. A first is algorithms that are designed to collude whenever they meet to play against a like-minded algorithm, yet without human coordination being required. This type we refer to as 'tacit collusion by algorithm'. A second type is algorithms that are used to implement a joint agreement in restraint of trade, which we refer to as 'explicit collusion by algorithm'.

3 Model and algorithm

To analyze how the observed phenomenon of supra-competitive limit prices when firms commit to Q-learning holds up to these criteria for algorithmic collusion, we closely examine a tractable version of Calvano et al. (2020)'s model, with just two feasible prices, the inner workings of which we first explain in detail. This simplest setting allows us to characterize the existence and nature of collusive equilibria, defined as strategy profiles that can be interpreted as having a reward-punishment scheme. The inference is that in more complex versions, such as the model with fifteen prices in Calvano et al. (2020), the patterns are likely to be less, and not more pronounced.[6]

Considered is a market environment with two competing firms, labelled 1 and -1, selling substitute products, each firm one. Time is discrete and indexed by $t \in \mathbb{N}$. At the beginning of each time period $t \in \mathbb{N}$, each firm $i = \pm 1$ selects an action (its selling price) $p_i(t) \in \mathcal{A}_i$ from a discrete, non-empty, and finite set of feasible prices $\mathcal{A}_i \subset [0, \infty)$. Subsequently, each firm observes their own demand $d_i(p_i(t), p_{-i}(t))$, where $d_i : [0, \infty)^2 \to [0, \infty)$ is a function unknown to the firms, and earns instantaneous reward $\pi_i(p_i(t), p_{-i}(t))$, where $\pi_i(p_i, p_{-i}) := (p_i - c_i) \cdot d_i(p_i, p_{-i})$ and $c_i \geq 0$ denote marginal costs. There are no demand shocks. The demand functions $d_i(\cdot, \cdot)$ are according to a multinomial logit model. That is, there are parameters $(a_{-1}, a_1, \mu) \in \mathbb{R}^2 \times (0, \infty)$, unknown to the firms, such that for both $i = \pm 1$,

$$d_i(p_i, p_{-i}) = \frac{\mathrm{e}^{\frac{a_i - p_i}{\mu}}}{1 + \mathrm{e}^{\frac{a_1 - p_1}{\mu}} + \mathrm{e}^{\frac{a_{-1} - p_{-1}}{\mu}}}, \tag{1}$$

for all nonnegative price pairs.[7]

Each firm's objective is assumed to be maximizing its cumulative expected discounted payoff stream,

$$\mathbb{E}\left[\sum_{t=1}^{\infty} \delta^t \pi_i(p_1(t), p_2(t))\right], \tag{2}$$

where $\delta \in (0, 1)$ is a common discount factor.

3.1 Description of the algorithm

Both firms use a Q-learning algorithm to determine the prices of their products from the action space. Essentially, Q-learning keeps track of a state variable $s_i(t)$ which consists of the prices charged in the previous time period: $s_i(t) = (p_i(t-1), p_{-i}(t-1))$, for all $t \in \mathbb{N}_{\geq 2}$.[8] The corresponding (finite) state space is equal to $\mathcal{S}_i := \mathcal{A}_i \times \mathcal{A}_{-i}$, for $i = \pm 1$. At the end of each time period $t \in \mathbb{N}$ (i.e., after rewards have been observed), the Q-learning algorithm of firm i computes a so-called Q-matrix $Q^{(i)}(t) = \{Q^{(i)}_{s,p}(t)\}_{s \in \mathcal{S}_i, p \in \mathcal{A}_i}$ via the recursion

$$Q^{(i)}_{s,p}(t) = Q^{(i)}_{s,p}(t-1) \text{ for all } (s,p) \in \mathcal{S}_i \times \mathcal{A}_i \text{ with } (s, p_i) \neq (s_i(t), p_i(t)),$$

[6]This inference is supported by an analysis of the three price setting in Meylahn (2023).

[7]Calvano et al. (2020) replace the 1 in the denominator of (1) by $e^{a_0/\mu}$. But since the demand function does not change if a constant is added to (a_{-1}, a_0, a_1), we can assume without loss of generality that $a_0 = 0$.

[8]Calvano et al. (2020) formulate a model with states consisting of the k most recent prices, but most results in the article and literature since then focus on the case $k = 1$.

and

$$Q^{(i)}_{s_i(t),p_i(t)}(t) = (1-\alpha)Q^{(i)}_{s_i(t),p_i(t)}(t-1) + \alpha\Big[\pi_i(p_i(t),p_{-i}(t)) + \delta \max_{p\in\mathcal{A}_\mathbf{i}}\{Q^{(i)}_{s_i(t+1),p}(t-1)\}\Big].$$

Here $\alpha \in [0,1]$ is a parameter called the *learning rate* and

$$Q^{(i)}(0) = \{Q^{(i)}_{s,p}(0)\}_{s\in\mathcal{S}_i, p\in\mathcal{A}_i}$$

is an initial Q-matrix. Both α and $Q^{(i)}(0)$ are input to the algorithm. $Q^{(i)}_{s,p}(t)$ is the Q-value corresponding to state s and price p. The Q-matrices are used to determine the prices. We assume that both firms employ ϵ_t*-greedy* action selection with $\epsilon_t := \exp(-\beta t)$ for all $t \in \mathbb{N}$ and some $\beta \geq 0$. This means that, for each $t \in \mathbb{N}$ and $i = \pm 1$, price $p_i(t)$ is selected uniformly at random from $\mathcal{A}_i$ with probability ϵ_t; with probability $1-\epsilon_t$, price $p_i(t)$ is selected in order to maximize the Q-value for the current state:

$$p_i(t) \in \arg\max_{p\in\mathcal{A}_i} Q^{(i)}_{s_i(t),p}(t-1),$$

with ties broken uniformly at random. The parameter β is called the *exploration rate*.

3.2 Market environment

Numerical experiments are conducted in a 'baseline parametrization' where marginal costs are $c_i = 1$ for both firms, and where the demand parameters are set to $(a_{-1}, a_1, \mu) = (2, 2, 1/4)$. The common discount factor is $\delta = 0.95$.

The feasible price sets are constructed as follows. First, the (unconstrained) Bertrand-Nash equilibrium prices (p_1^N, p_{-1}^N) of the one-shot pricing game, and the (unconstrained) prices (p_1^M, p_{-1}^M) that maximize joint profit, are computed. These prices are given by

$$(p_1^N, p_{-1}^N) := \left(\frac{\mu}{1 - V(\mathrm{e}^{(a_1-c_1)/\mu-1}\tilde{Q}_0)} + c_1, \frac{\mu}{1 - V(\mathrm{e}^{(a_{-1}-c_{-1})/\mu-1}\tilde{Q}_0)} + c_{-1}\right),$$

$$(p_1^M, p_{-1}^M) := \Big(\mu(1+W(A_0)) + c_1, \mu(1+W(A_0)) + c_1\Big),$$

where: $V(x)$ is the unique solution v in $(0,1)$ of $v \cdot \exp(v/(1-v)) = x$, $\tilde{Q}_0$ is the unique solution Q_0 to $Q_0 + V(Q_0\mathrm{e}^{(a_1-c_1)/\mu}) + V(Q_0\mathrm{e}^{(a_{-1}-c_{-1})/\mu}) = 1$, $W(x)$ is the Lambert W function which solves $W(x)\cdot \mathrm{e}^{W(x)} = x$, and $A_0 := \exp((a_1 - c_1)/\mu - 1) + \exp((a_{-1} - c_{-1})/\mu - 1)$.[9]

The baseline parametrization is fully symmetric: $c_1 = c_{-1}$ and $a_1 = a_{-1}$, resulting in Nash prices $p^N := p_1^N = p_{-1}^N = 1.4729$ with corresponding payoff $\pi^N := \pi_i(p_i^N, p_{-i}^N) = 0.2229$, and monopoly prices $p^M := p_1^M = p_{-1}^M = 1.9250$, with corresponding payoff $\pi^M := \pi_i(p_i^M, p_{-i}^M) = 0.3375$. Hence, in perfect collusion, in which both firms charge the monopoly price, the cartel increases

[9] For a derivation of these expressions, we refer to Li and Huh (2011), Theorem 4 and Theorem 2, respectively.

profit by 51 percent. Note that there is positive profit in competition due to the assumption of a logit demand function.

For some integer $m \geq 2$ and some $\xi \in [0, \max_{i=\pm 1} \frac{p_i^N}{p_i^M - p_i^N}]$, the feasible price sets are defined by

$$\mathcal{A}_i := \left\{ p_i^N - \xi(p_i^M - p_i^N) + k \cdot (1 + 2\xi) \frac{(p_i^M - p_i^N)}{m - 1} : k = 0, \ldots, m - 1 \right\},$$

for both players $i = \pm 1$. For $m = 2$ and $\xi = 0$ this results, for example, in the feasible price set $\mathcal{A}_i = \{p_i^N, p_i^M\}$. For higher values of ξ, the firms get more pricing options below the competitive and above the monopoly price. Calvano et al. (2020)'s $m = 15$ and $\xi = 0.1$, for both players, result in the feasible price set

$$\mathcal{A}_i := \{1.4277, 1.4277 + 0.0387, 1.4277 + 2 \times 0.0387, \ldots, 1.9702\},$$

so that the lowest feasible price is slightly (3 percent) below the Nash price and the highest is slightly (2 percent) above the monopoly price.

The initial Q-matrix for both firms $i = \pm 1$ is set to

$$Q_{s,p}^{(i)}(0) = \frac{1}{(1 - \delta)} \frac{1}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \pi_i(p_i, p_{-i}).$$

Numerical experiments are conducted for all possible pairs

$$\begin{aligned}(\alpha, \beta) \in \Psi := &\{\alpha_{\min} + \frac{i \cdot (\alpha_{\max} - \alpha_{\min})}{99} : i = 0, 1, \ldots, 99\} \\ &\times \{\beta_{\min} + \frac{j \cdot (\beta_{\max} - \beta_{\min})}{99} : j = 0, 1, \ldots, 99\},\end{aligned}$$

with $\alpha_{\min} = 0.025$, $\alpha_{\max} = 0.25$, $\beta_{\min} = 0$, and $\beta_{\max} = 2 \times 10^{-5}$. For each $(\alpha, \beta) \in \Psi$, 1000 simulations are conducted in which the two Q-learning algorithms play against each other.

4 Limitations of pricing by Q-learning algorithms

To develop our main points, and due to the interdependence of the arguments, we start by analyzing the performance of Q-learning with respect to the firm's stated objective in Section 4.1. We subsequently show in Section 4.2 that the process by which the algorithms learn collusive strategies is slow relative to the timescale imposed by the firm's objective. In sections 4.3 and 4.4, we examine the strategies learned by the algorithms and whether pricing patterns can be used to infer properties of the learned strategies, respectively. Finally, we evaluate the performance of Q-learning against a simple alternative algorithm in Section 4.5. All simulations are performed using Q-learning in the market environment of Calvano et al. (2020), which we validate by replicating their key Figure 3 in Appendix A.

4.1 Performance measure and short-term gains

To evaluate the present discounted value for firms of using a Q-learning pricing algorithm, we first derive a natural timescale imposed by the firm's objective, and then show that supra-competitive profits earned by the firms on this timescale are the result of the chosen action space.

4.1.1 Extra-profits upon convergence have negligible value

Given the firms' assumed objective, it is reasonable to evaluate the performance of the pricing algorithms in terms of the cumulative expected discounted profits they generate. In the context of (algorithmic) collusion, it would be natural to compare the cumulative expected discounted earnings obtained under the pricing algorithms with what would have been earned under the Nash prices and monopoly prices, and measure the (firms' benefits of) collusion, for example, by

$$\begin{aligned}\tilde{\Delta} &:= \frac{\mathbb{E}\left[\sum_{t=1}^{\infty}\delta^t\pi_i(p_i(t),p_{-i}(t))\right]-\sum_{t=1}^{\infty}\delta^t\pi^N}{\sum_{t=1}^{\infty}\delta^t\pi^M-\sum_{t=1}^{\infty}\delta^t\pi^N}\\ &= \frac{(1-\delta)\mathbb{E}\left[\sum_{t=1}^{\infty}\delta^{t-1}\pi_i(p_i(t),p_{-i}(t))\right]-\pi^N}{\pi^M-\pi^N}.\end{aligned}$$

Calvano et al. (2020), and most of the studies since, consider the firms to be maximizing cumulative expected discounted profits, yet use Δ, a normalized 'average per-firm profit' compared to the static Nash equilibrium *upon convergence*, as a performance measure for the benefits of collusion.[10] Formally it is defined as

$$\Delta := \frac{\bar{\pi}-\pi^N}{\pi^M-\pi^N},$$

where $\bar{\pi}$ is the 'average per-firm profit upon convergence'.[11]

Based on this performance metric, Q-learning generates a sizable extra profit. High values of Δ are not an indication, however, that the firms benefit from using the algorithms in terms of their actual objective. Because it may take a long time before supra-competitive prices from collusion arise, the actual contribution of this profit gain to the firms' stated objective of cumulative discounted profit may be negligible.

[10] In Calvano et al. (2020), a simulation is said to converge if there is a $t_0 \in \mathbb{N}$ with $t_0 \leq 10^9 - 10^5$ and actions $\{a_i(s) \in \mathcal{A}_i : i = \pm 1, s \in \mathcal{S}_i\}$ such that $a_i(s) = \arg\max_{p\in\mathcal{A}_i} Q^{(i)}_{s,p}(t)$ for all $t = t_0+1, \ldots, t_0+10^5$ and all $i = \pm 1$, $s \in \mathcal{S}_i$. That is, there is convergence if the optimal actions corresponding to all firms and states are unique and remain constant during 100,000 consecutive time periods, within the first billion time periods.

[11] From the description in Calvano et al. (2020) (p. 3277) we interpret $\bar{\pi}$ to be the *observed* average profit over the time periods $t_0+1, \ldots, t_0+100,000$. Note furthermore that this performance metric assumes a symmetric demand function. Note also that the Nash and monopoly prices p_i^N and p_i^M are not necessarily included in the feasible price sets, so that $\Delta \in \{0, 1\}$ is not necessarily attainable by any policy. The metric Δ is not defined in case a simulation does not converge (in the sense of Calvano et al. (2020)), but this rarely occurs in our numerical experiments.

To make the latter concrete, define an *effective time horizon*

$$T_\delta := \left\lceil \frac{\log(\frac{\pi_{\min}}{1000\pi_{\max}})}{\log(\delta)} \right\rceil,$$

where $\pi_{\min}$ and $\pi_{\max}$ are the smallest and largest value in the pay-off matrix.[12] The payoffs obtained after T_δ time periods contribute less than 0.1 percent to the overall profit, and thus are almost irrelevant from the firm's point of view. In the base parametrization, $\pi_{\min} = 0.0911$ and $\pi_{\max} = 0.4270$, so that $T_\delta = 165$ for $\delta = 0.95$. Thus, from the firm's point of view, all profits obtained in the baseline parametrization after the first 165 time periods are practically irrelevant, let alone what happens after 400,000 time periods, which is the time after which Calvano et al. (2020) report convergence. Note that the number of Q-values each algorithm needs to learn is 15^3 in the baseline parameterization, meaning that a maximum of 5% of the Q-values will have been updated during the effective time horizon.

Two alternative interpretations of the results presented in Calvano et al. (2020) may circumvent the requirement of an algorithm to learn optimal Q-values for significantly more state-action pairs than samples available in the effective time horizon, which crucially depends on δ. (1) the discount rate δ may be interpreted as an algorithmic hyperparameter that needs to be tuned to maximize the firm's true objective, which may make use of a second discount factor or may be based on the expected average reward, but this would require making the true objective of the firm explicit and evaluating the performance based on it. (2) the intended implementation of the algorithms may be that they have been trained offline before being implemented online. In that case, *all* evaluation should be carried out by using the firm's stated objective and for algorithms trained in two independent copies of the same environment.[13] Experiments of the latter kind are conducted in (Calvano et al., 2020, Figure 11), but the algorithms are only evaluated in terms of Δ as a function of time, so we cannot know how well they do in present discounted value terms. More extensive experiments of this kind are conducted by Eschenbaum, Mellgren and Zahn (2022). However, these still do not entirely eliminate the need for coordination, as the algorithms are, for example, assumed to have the same hyperparameters, start at the same time, and/or have the same state and action spaces.

4.1.2 Short-run gains are not caused by collusion but choice of the feasible price set

The Q-learning algorithm in Calvano et al. (2020) sometimes generates supra-competitive profits within the firms' effective time horizon. This is not caused by collusion, but by the fact that, for small enough values of δ, including $\delta = 0.95$, the algorithm behaves nearly identically to pricing uniformly at random within the effective time horizon.

To see this, note that the probability that the Q-learning algorithm does *not* experiment but

[12] The expression for T_δ is obtained by considering the worst-case setting where the firm earns the lowest possible payoff $\pi_{\min}$ until time T_δ and earns the highest possible payoff $\pi_{\max}$ thereafter, and then solving for the value of T_δ such that the contribution to the firms objective after T_δ is 0.1 percent.

[13] The reason for this is that evaluating algorithms trained under self-play after convergence would correspond to an exchange of or agreement on Q-value tables between firms after offline training in a real-world setting.

prices according to the Q-matrices within the effective time horizon, is bounded from above by $1 - \exp(-\beta_{\max} T_\delta) = 1 - \exp(-2 \times 10^{-5} \times 165) \approx 0.003295$. The algorithm therefore prices uniformly at random in, on average, more than 164 of the 165 time periods.[14]

In the base parametrization, the demand model parameters and feasible price set are such that, if both firms price uniformly at random, prices are on average 1.6990 and thus higher than the Nash price 1.4729. The corresponding average per-period profits are 0.2799, larger than the per-period Nash profits of 0.2229. This profit gain disappears under different choices of the feasible price set.

Suppose, for example, that the price set is bounded from below by the marginal cost and is symmetric around the Nash price:

$$\hat{\mathcal{A}}_i = \Big\{ c_i + \frac{k}{m+1} \cdot 2(p_i^N - c_i) : k = 1, \dots, m \Big\}.$$

The average price under pricing-uniformly-at-random is then exactly equal to the Nash price, and the average per-period profit when both players price uniformly at random equals 0.17228, which is 22 percent *smaller* than the Nash profits.[15] Any supra-competitive gains to the firms' actual objective are caused by the performance of pricing-uniformly-at-random, and not by collusive behavior.

4.2 Convergence is intrinsically slow

In this section, we argue that the choice of state space ensures that it is possible for collusive strategy equilibria to exist (Section 4.2.1), that their existence changes dynamically and is such that they only exist after the effective time horizon (Section 4.2.2), and that changing the algorithmic hyperparameters to shift their time of existence into the effective time horizon drastically reduces the probability of learning collusive strategy equilibria (Section 4.2.3).

4.2.1 Conditions for the existence of collusive strategy equilibria

The supra-competitive limit prices alone, Calvano et al. (2020) do not take as per se proof of collusion. In addition, following Harrington Jr. (2018), the authors require for collusion the presence of a reward-punishment scheme designed to provide the incentives for firms to consistently price above the competitive level. This requirement is crucial to them, as: 'The reward-punishment scheme ensures that the supra-competitive outcomes may be obtained *in equilibrium* and do not result from a failure to optimize.'[16]

The possibility that pricing based on Q-learning converges to collusive equilibria only exist,

[14] This changes if δ approaches one or β grows large, but this is problematic for reasons discussed in Section 4.2.3.

[15] Changing the parameter ξ, which controls the distance between consecutive feasible prices, also affects the average price and profit during the effective time horizon: increasing ξ from 0.1 to 1.0, for example, reduces the per-period profit to 0.1916, or 14 percent *below* the Nash per-period profits in case both firms price uniformly at random around the Nash-price.

[16] Calvano et al. (2020), p. 3269. This would contrast their work with Waltman and Kaymak (2015) and Cooper, Homem-de Mello and Kleywegt (2015), which would be 'failure to learn' (p. 3269) and 'collusion by mistake' (p. 3269), respectively.

however, by construction of the state space. To see how, consider first the system where both players use Q-learning with *fixed* exploration probability $\epsilon \in [0,1]$. Let $\mathcal{A}_i$ denote the feasible price set of firm $i = \pm 1$, let a mapping from the state space $\mathcal{A}_i \times \mathcal{A}_{-i}$ to $\mathcal{A}_i$ be called a *strategy* of player i, and let Σ_i denote the collection of all such mappings. A strategy $\sigma^{(i)} = \{\sigma^{(i)}(s) : s \in \mathcal{A}_i \times \mathcal{A}_{-i}\}$ of player i is called (δ, ϵ)-*best-response* to strategy $\sigma^{(-i)} = \{\sigma^{(-i)}(s) : s \in \mathcal{A}_{-i} \times \mathcal{A}_i\}$ of player $-i$ if, for all states $s = (s_i, s_{-i}) \in \mathcal{A}_i \times \mathcal{A}_{-i}$,

$$\begin{aligned}\sigma^{(i)}(s) \in \arg\max_{p \in \mathcal{A}_i} \frac{\epsilon}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} &\{\pi_i(p, p_{-i}) + \delta V^{(i)}_{\sigma^{(-i)}}(p, p_{-i})\} \\ &+ (1-\epsilon)\{\pi_i(p, \sigma^{(-i)}(s_{-i}, s_i)) + \delta V^{(i)}_{\sigma^{(-i)}}(p, \sigma^{(-i)}(s_{-i}, s_i))\}, \end{aligned} \tag{3}$$

where, for all states $s = (s_i, s_{-i}) \in \mathcal{A}_i \times \mathcal{A}_{-i}$,

$$\begin{aligned}V^{(i)}_{\sigma^{(-i)}}(s) := \max_{p \in \mathcal{A}_i} \frac{\epsilon}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} &\{\pi_i(p, p_{-i}) + \delta V^{(i)}_{\sigma^{(-i)}}(p, p_{-i})\} \\ &+ (1-\epsilon)\{\pi_i(p, \sigma^{(-i)}(s_{-i}, s_i)) + \delta V^{(i)}_{\sigma^{(-i)}}(p, \sigma^{(-i)}(s_{-i}, s_i))\}. \end{aligned} \tag{4}$$

Equation (4) defines the value function for player i when the opponent plays according to $\sigma^{(-i)}$ with probability $1 - \epsilon$ and selects an action uniformly at random with probability ϵ. Equation (3) defines the corresponding optimal actions.[17] A strategy profile $(\sigma^{(i)}, \sigma^{(-i)}) \in \Sigma_i \times \Sigma_{-i}$ is called a (δ, ϵ)-*strategy equilibrium* if the strategies are (δ, ϵ)-best-responses to each other.

To gain insight into the existence of and convergence to (δ, ϵ)-equilibria that can be called 'collusive', consider the setting with $m = 2$ feasible prices for each firm: the Nash price p^N and the monopoly price p^M. Since $m = 2$ corresponds to only 256 possible strategy profiles, this allows us to compute all (δ, ϵ)-strategy equilibria by brute force enumeration, for any given feasible values of δ and ϵ.[18] Using the usual notation for the prisoner's dilemma, write $C := p^M = 1.9250$ for the action 'cooperate', $D := p^N = 1.4729$ for the action 'defect', and write $R := \pi_i(C, C) = 0.3375$, $S := \pi_i(C, D) = 0.1180$, $T := \pi_i(D, C) = 0.3679$, $P := \pi_i(D, D) = 0.2229$ for the corresponding payoffs in the baseline parametrization. There are 16 possible strategies $\{CCCC, CCCD, CCDC, CCDD, \ldots, DDDD\}$, where the four consecutive letters of a strategy refer to the actions taken in state (C, C), (C, D), (D, C), and (D, D), in that order. For each $\delta \in (0, 1)$ and $\epsilon \in [0, 1]$, we have computed all possible (δ, ϵ)-strategy equilibria by self-consistently solving the Bellman equations (3) and (4).[19] These are reported in Table 1 and visualized in Figure 1.

These computations reveal that there are three strategy equilibria: both players using 'All Defect' (AD, which always plays D), 'Grim Trigger' (GT, which plays C in state (C, C), and D otherwise)[20], or 'Win-Stay, Lose-Shift' (WSLS, which plays C in states (C, C) and (D, D), and

[17]These value functions incorporate random exploration of the competing firm, but not the firm's own random explorations, reflecting the fact that Q-values are updated after determining the price.

[18]For general m there are m^{2m^2} strategy profiles, which for $m = 3$ already equals 387,420,489 (analyzed in Meylahn (2023)), and for $m = 15$, which Calvano et al. (2020) consider, is a number with 530 digits.

[19]For details of these computations, see Meylahn and Janssen (2022); Barfuss and Meylahn (2023); Meylahn (2025).

[20]Note that this is a one-period memory version of the usual Grim Trigger strategy. In particular, cooperation

D otherwise). The AD equilibrium exists for all (ϵ, δ), but is clearly not collusive. The strategies WSLS and GT can be interpreted as having a reward-punishment scheme, as they play C in state (C, C) ('rewarding' the competitor playing C) and play D in state (C, D) ('punishing' the competitor who deviates from C). The WSLS equilibrium exists for all points (ϵ, δ) lying above the solid line in Figure 1, and GT exists for the points (ϵ, δ) lying between the two dashed curves (the upper dashed curve is barely visible in the top left corner) in Figure 1. Figure 1 also reveals that WSLS or GT can only be collusive equilibria if ϵ is sufficiently small. The existence of collusive strategy equilibria is pre-programmed, as it were, in the dynamical system by the inclusion of a memory of one period.[21]

Strategy of both players	Conditions for equilibrium
DDDD ('All Defect')	No additional conditions
CDDD ('Grim Trigger')	$\frac{\epsilon(S+T-R-P)+2(R-T)}{(\epsilon^2-3\epsilon+2)(P-T)} < \delta < \frac{\epsilon(S+T-R-P)+2(P-S)}{(1-\epsilon)(\epsilon(T-P)+2(P-S))}$
CDDC ('Win-Stay, Lose-Shift')	$\delta > \frac{\epsilon(P+R-S-T)+2(T-R)}{(1-\epsilon)(\epsilon(P+S-T-R)+2(R-P))}$

Table 1: Conditions on $\delta \in (0, 1)$ and $\epsilon \in (0, 1)$ such that the strategies in the left column form (δ, ϵ)-strategy equilibria if used by both players.

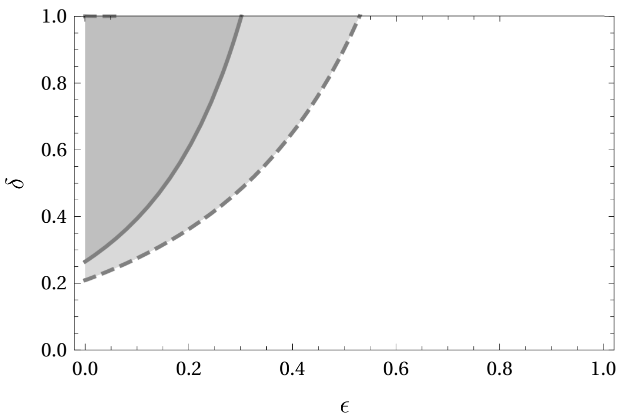

Figure 1: Phase diagram of (δ, ϵ)-strategy equilibria. AD is always an equilibrium, GT is an equilibrium if (δ, ϵ) lies above the large dashed curve and below the tiny dashed curve in the upper left corner (light and dark gray areas), and WSLS is an equilibrium if (δ, ϵ) lies above the solid curve (dark gray area).

4.2.2 Collusive equilibria only exist after the effective time horizon

The speed with which the algorithms converge depends on the decreasing exploration rate in two ways: (1) there is an initial period in which the collusive strategy equilibria do not exist yet, and (2) after they exist, a specific sequence of events has to occur in order to transition to a collusive strategy equilibrium. Supra-competitive limit prices sustained by collusive strategy equilibria can only be learned after a sufficiently long exploration period because only then do they come into existence. To see this, consider the extreme case of a fixed value $\epsilon = 1$, for which we know there are no collusive equilibria—compare Figure 1. In that case, all immediate

can be restored if both players explore and play C.

[21] The memoryless version analyzed by Dolgopolov (2024) and Lambin (2024) would be an example of a state space choice that does not pre-program collusive strategy equilibria in this sense.

payoffs in (4) are state-independent, from which it follows that the value function is state-independent—and equal to $V^{(i)}_{\sigma(-i)}(s) = (1-\delta)^{-1} \max_{p\in\mathcal{A}_i} |\mathcal{A}_{-i}|^{-1} \sum_{p_{-i}\in\mathcal{A}_{-i}} \pi_i(p, p_{-i})$ for all states s. In the baseline parametrization, there is a unique price $p^* = 1.58271$ at which the optimum in this maximization problem is attained, and the strategy σ^* defined by $\sigma^*(s) = p^*$, for all states s, constitutes a $(\delta, 1)$-strategy equilibrium (σ^*, σ^*). Because this strategy always sets the optimal price assuming that the opponent plays uniformly at random, and therefore does not contain any structure that could be interpreted as 'reward' or 'punishment', this strategy should be considered non-collusive, i.e., 'competitive', and the equilibrium (σ^*, σ^*) a non-collusive equilibrium of strategies.[22] By continuity properties of $V^{(i)}_{\sigma(-i)}(s)$, considered as a function of ϵ, it follows that this equilibrium is the only one possible if ϵ is sufficiently large. This is a general result, which holds for any number of prices, that we formulate as a theorem.

Theorem 1. *For all $\delta \in (0, 1)$ there is an $\epsilon(\delta) \in [0, 1)$ such that, for all $\epsilon \in (\epsilon(\delta), 1]$, the only (δ, ϵ)-strategy equilibrium is the non-collusive equilibrium (σ^*, σ^*).*

Proof. Proof. See Appendix G. □

Q-learning with *time-dependent* exploration probability $\epsilon_t = \exp(-\beta t)$ does not admit stationary equilibria in the sense defined in Section 4.2.1. The underlying dynamical system can nevertheless be considered as a system for which the attracting equilibria at time t are precisely the (δ, ϵ_t)-strategy equilibria. So considered, Figure 1 suggests that convergence to a collusive strategy equilibrium is *not possible* before ϵ_t has dropped below a critical value $\epsilon(\delta)$, and thus is not possible before a critical time period $T_{\epsilon(\delta),\beta} := \lfloor -\log(\epsilon(\delta))/\beta \rfloor$. Because Calvano et al. (2020) choose β very small, this critical time period will be large. This explains, in part, why the authors find very slow convergence: convergence to a collusive strategy equilibrium *can* only start to happen after a very long time, when ϵ_t has declined sufficiently for collusive strategy equilibria to exist.

To validate this reasoning, we simulated two independent Q-learning algorithms with $\delta = 0.95$ and $\alpha = 0.15$, conform the representative experiment in Calvano et al. (2020).[23] We set $\beta = 10^{-4}$, which is lower than the representative value 4×10^{-6}, because with two prices less exploration is needed than with fifteen prices.[24] It follows from Table 1 that GT is an equilibrium strategy for all $\epsilon < 0.515126$, and WSLS is an equilibrium strategy for all $\epsilon < 0.292042$. This means that for all $t < 6633$ only AD is an equilibrium strategy, for $6633 < t < 12308$ both AD and GT are equilibrium strategies, and for $t > 12308$ all three of AD, GT, and WSLS are equilibrium strategies. Based on these computations, we expect that strategies will not converge to GT before $t = 6633$ and will not converge to WSLS before $t = 12308$.

These expectations are confirmed by the outcomes of our numerical experiments. The left panel of Figure 2 shows, based on 1000 simulations, how many trajectories are playing the strategy profiles AD, GT, WSLS, or something else, as a function of time. Before $t = 6633$ (the first

[22] This does not mean that Δ will be small; but as explained in Appendix F, this is simply a consequence of how the feasible price set is constructed.

[23] Table 3 in Appendix C shows that other choices of α lead to qualitatively similar results.

[24] Table 4 in Appendix D shows that alternative choices lead to qualitatively similar findings.

vertical dashed line) the fraction of trajectories in the AD strategy profile is steadily increasing (after an initial drop caused by the initialization). At $t = 6633$ the fraction of GT strategy profiles slowly starts increasing, and only at time $t = 12308$ (the second vertical dashed line) WSLS is learned—before that, the fraction of sample paths in which both firms play WSLS is negligible. Thus, we see that the structure of (δ, ϵ) strategy equilibria as defined by the Bellman equations (3) and (4), combined with the fact that ϵ_t decreases from one to zero, gives rise to a *phase transition* that makes convergence to collusive equilibria possible.

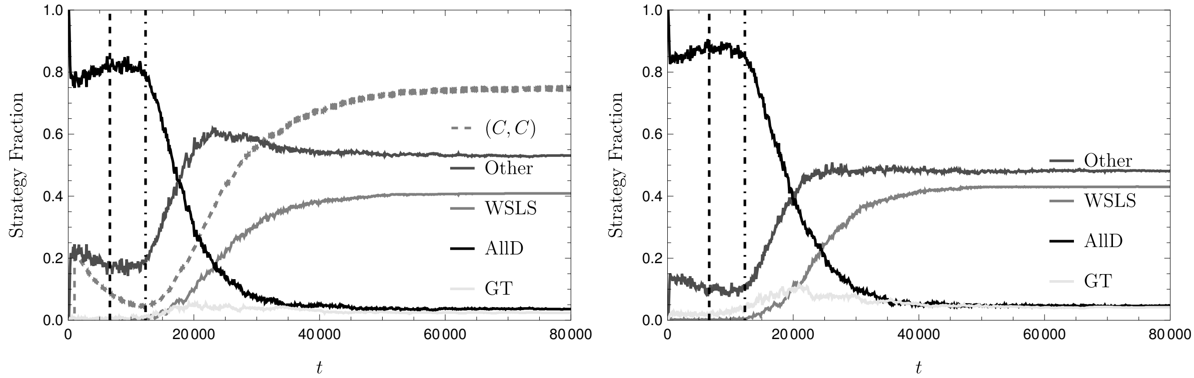

Figure 2: Left: Fraction of trajectories in the AD strategy profile (black), GT strategy profile (Light Gray), WSLS strategy profile (Gray) and other strategy profiles (Dark Gray). Also plotted is the fraction of periods spent in the (C, C) state (Gray dashed) in the 1000 most recent time periods. The dashed line indicates the time at which the GT strategy profile becomes stable, and the horizontal dash-dotted line indicates the point at which the WSLS strategy profile becomes stable. Right: A simulation with the same parameters, but plotting the strategy fractions for player one only.

Note further that from $t = 12308$ to $t \approx 40000$, in the left panel of Figure 2, 41 percent of the trajectories start converging to WSLS. The other equilibria eventually attract 3.6 percent (AD) and 2.4 percent (GT) of the samples. The convergence is remarkably slow for a number of reasons, related to the inner dynamics of ϵ_t-greedy Q-learning. To understand these, let $Q^{(i)}_{s,p}(t)$ denote the Q-value of player i for state s and price p at time t, and consider the strategy equilibrium WSLS. In order to reach this strategy equilibrium, it is necessary that action C becomes the optimal action in state (C, C) for both players, i.e., that $Q^{(i)}_{(C,C),C}(t) > Q^{(i)}_{(C,C),D}(t)$ for both $i = \pm 1$. Before $t = 12308$, the majority of optimal strategies according to the Bellman equation (3) are AD, so that, in these cases, $\Delta Q^{(i)}_{(C,C)}(t) := Q^{(i)}_{(C,C),C}(t) - Q^{(i)}_{(C,C),D}(t) < 0$ and state (C, C) can only be reached if both players simultaneously explore and select action C—this happens with probability $\epsilon_t^2/4$, which ranges from 0.02132 at time 12308 to just 0.00008387 at time 40000. Being in state (C, C), $\Delta Q^{(i)}_{(C,C)}(t)$ can increase if both players play C (so that $Q^{(i)}_{(C,C),C}(t)$ is increased after updating)—this initially only happens by random exploration, i.e., with probability $\epsilon_t^2/4$—or if both players play D (and $Q^{(i)}_{(C,C),D}(t)$ is decreased after updating). However, in state (C, C), it might also happen that $\Delta Q^{(i)}_{(C,C)}(t)$ *decreases*: namely if player i plays C (e.g., by random exploration) and player $-i$ plays D (so that player i receives the S payoff and $Q^{(i)}_{(C,C),C}(t)$ is decreased after updating), or, conversely, if player $-i$ plays C and player i plays D (so that player i receives the T payoff and $Q^{(i)}_{(C,C),D}(t)$ is increased after updating). Thus, it might take several attempts of reaching state (C, C) by simultaneous random exploration before $\Delta Q^{(i)}_{(C,C)}(t)$ becomes positive.

An additional challenge is that the moments that this happens for both players should not be too far apart: if $\Delta Q^{(i)}_{(C,C)}(t) > 0$ but $\Delta Q^{(-i)}_{(C,C)}(t) < 0$, then, if both players do not explore (which gets increasingly more likely as time increases), player i will receive the payoff S which will decrease $Q^{(i)}_{(C,C)}(t)$ and might make $\Delta Q^{(i)}_{(C,C)}(t)$ negative again. In fact, even if $\Delta Q^{(i)}_{(C,C)}(t) > 0$ for both players $i = \pm 1$, a series of random explorations in which the players do not play the same action in state (C, C) can again reverse the sign of $\Delta Q^{(i)}_{(C,C)}(t)$ for one of the players. This explains why it may take a considerable time to converge to the WSLS strategy profile, and why, when ϵ_t has gotten too small, it becomes increasingly unlikely that strategy profiles different from WSLS will converge to WSLS.

Hence, there are two causes why convergence to collusive strategy profiles is slow and partial. First, collusive strategy equilibria only exist, and thus convergence to them is only possible, if ϵ_t is sufficiently small (and δ sufficiently large). Because β is small, it takes a large number of time periods before the required phase transition occurs. Second, if collusive strategy equilibria exist—i.e., ϵ_t and δ satisfy the required conditions—it requires a certain number of random events to happen within a particular time frame to reach a collusive equilibrium, which can fail to materialize or take a considerable amount of time.

4.2.3 Convergence cannot easily be sped up

The prohibitively slow and partial convergence to collusive outcomes cannot be fixed by ad-hoc modifications to the algorithm.[25] We discuss three natural candidates. First, one could change the exploration rate $\epsilon_t = \exp(-\beta t)$. Three ideas are (i) set $\epsilon_t = \epsilon_0 \exp(-\beta t)$ with $\epsilon_0 := 0.292042$, so that convergence to WSLS is immediately possible instead of only at $t = 12308$; (ii) increase β so that the time $\lceil -\log(\epsilon_0)/\beta \rceil$ until ϵ_0 is reached is much shorter; (iii) set a *fixed* exploration rate below the critical value ϵ_0. None of these approaches solves the problem, however: simulations indicate that in (i) convergence to WSLS still occurs far beyond the effective time horizon, while it reduces the fractions converging to WSLS from 41 to 36 percent; in (ii), increasing β reduces the amount of collusion (only 16 percent of the samples converges to WSLS when $\beta = 10^{-3}$), while decreasing β increases the time of the phase transition; in (iii), $\epsilon_t \equiv 0.1$ reduces collusive limit strategy profiles to less than 20 percent. For the details of these simulations, see Appendix D.

Second, one may set the discount factor δ very close to one, in order to increase the effective time horizon T_δ. Figure 3 in Calvano et al. (2020) appears to suggest that choosing δ close to one can only be beneficial for collusion.[26] However, their performance measure Δ does not measure collusion, as whether or not the strategies underlying an equilibrium of limit prices involve reward-punishment schemes cannot be inferred from high values of Δ. The high values

[25]More advanced learning techniques, such as Deep Learning Hettich (2021), may speed up learning, but the results of Hettich (2021) show that collusion is still only learned after the effective time horizon, and the chosen exploration rate function again prohibitively delays the existence of collusive equilibria.

[26]But see Calvano et al. (2020), footnote 28, where the authors point to a failure of Q-learning when δ is very close to one.

around 70 to 90 percent are partly caused by the choice of the feasible price set. In particular, choosing the fifteen prices *symmetric* around the Nash price makes Δ decrease as δ approaches one. For our tractable $m = 2$ setting, we evaluated the *fraction of collusive limit strategy equilibria* in the baseline parametrization (with $m = 2$) and observed that this fraction actually converges to zero as δ approaches one.[27] For the details, see Appendix E.

Third, offline training could be used to speed up convergence. However, contrary to AlphaGo or other board games, pricing games crucially involve unknown market parameters that can only be learned through real interactions with actual demand. Calvano et al. (2020) acknowledge that offline learning may be of little help in colluding actually, yet nevertheless suggest that numerical offline experiments could help overcome the problem of slow learning.[28] However, this conclusion is based on Δ in their Figure 11, which as noted does not measure collusion or profit gain according to the firms' actual objective. Concretely, the timescale of their Figure 11 is still significantly larger than the effective time horizon—compare Section 4.1.

4.3 Based on simulations, the majority of limit strategy profiles are not collusive

While pricing based on Q-learning converges remarkably often to supra-competitive prices in the long time limit, a substantial fraction of these are not sustained by collusive strategy equilibria. Our simulations for $m = 2$ prices show that Q-learning algorithms do not 'systematically learn to play collusive strategies'. As shown in the left panel of Figure 2, about 57 percent of the simulations do not converge to a collusive strategy equilibrium (3.6 percent AD, 53.1 percent other). The collusive strategy equilibria thus account for about 43 percent, with 40.9 percent contributed by WSLS and 2.4 percent contributed by GT.[29] The right panel of Figure 2 shows that about 53 percent of the simulations do not converge to collusive single-player strategies (4.8 percent AD, 48.1 percent other). We thus observe in simulations that the majority of limit strategy profiles and limit strategies are not collusive.[30] In addition, the left panel of Figure 2 shows that the fraction of time that both players charge supra-competitive prices converges to about 74 percent. Since random exploration has almost vanished at this point, this implies that a substantial fraction (44.7 percent) of supra-competitive limit prices are *not* generated by collusive equilibria.

One might object that it is irrelevant whether supra-competitive prices are generated by collusive equilibria or strategies, and that all that matters is whether there is an underlying reward-

[27] This reduction can partially be countered by decreasing β, but this in turn comes at the cost of increasing the time of the phase transition.

[28] Calvano et al. (2020), p. 3294.

[29] The GT strategy equilibrium, which is a classic example of a reward-punishment scheme, only accounts for approximately 2 percent of the limiting strategy profiles. It, however, does not sustain supra-competitive prices, as the stationary distribution it induces on the states concentrates on the (D, D) state as the exploration rate vanishes.

[30] These findings are conceptually different from the observation reported in Table 1 in Calvano et al. (2020). There, approximately one half of simulations converged to a Nash equilibria on path (whether a strategy profile is deemed to be a Nash equilibrium is checked only for states actually played). Their Table 1 is about (near-)convergence to strategy equilibria, but does not report how often these limiting strategy equilibria were collusive—that is, contained a reward-punishment scheme. Figure 2 reveals that in our case, more than half of the limiting strategy profiles are not collusive.

punishment scheme. This would depart from Calvano et al. (2020) and subsequent research, for whom being in or near strategy equilibrium is important.[31] Let us nevertheless consider whether the supra-competitive limit prices can be explained by reward-punishment schemes. For $m = 2$ prices, call a strategy RP if it plays C in state (C, C) (the 'reward' of sustaining high prices) and D in state (C, D) (the 'punishment' of defecting from the high prices). There are four such strategies, and sixteen RP strategy profiles. It turns out that in 46 percent of the simulations, both players converge to an RP strategy. This is substantially lower than the 74 percent of the simulations with supra-competitive limit prices (C, C). Thus, resorting to the weaker notion of RP-strategies instead of collusive equilibria or collusive strategies does not offer a complete explanation of the observed supra-competitive limit prices. A significant fraction of supra-competitive prices is the result of incomplete learning, not learning to collude. As in the original work on incomplete learning, convergence often is to 'incorrect' prices, that correspond to 'subjective equilibria' that are different from equilibria of the underlying model.[32] Moreover, the remaining 40.9 percent observations of 'collusion' by this definition—which may seem still substantive and a concern for regulators—are of limited practical concern as autonomous algorithmic collusion. This is because the likelihood of this kind of algorithmic interaction to emerge spontaneously is negligible, as we argue in Section 4.5, the probability of both players using Q-learning in equilibrium is nearly zero.

4.4 What looks like collusion need not be collusion

Calvano et al. (2020) search for price patterns that indicate reward-punishment schemes in the limiting strategies, and take those as evidence of collusive equilibria: '[t]he reward-punishment scheme ensures that the supra-competitive outcomes may be obtained *in equilibrium* and do not result from a failure to optimize' (emphasis original).[33] Starting from any limiting strategy profile, the authors force one firm to cut its price during a single time period, and then count how often: (a) the forced price cut by one firm is followed by a price decrease by the other firm, (b) after a number of time periods the prices return to their original level, and (c) the total discounted profit of the deviating player is lower than if there would have been no deviation at all.

This approach is problematic, however, because the pattern (a,b,c) does not imply that the underlying strategies are collusive strategy equilibria, or have a reward-punishment structure.[34] To see this, with $m = 2$ prices, suppose that firm 1 plays the strategy $CCDC$ and firm -1 plays $CDCC$. This leads to the strategy graph shown in Figure 3, where nodes are labeled from the perspective of player 1. This strategy profile is different from GT and WSLS, so it is not a collusive strategy equilibrium. Suppose that the pre-shock price at time $\tau = 0$ is C for both

[31] This is what Calvano et al. (2020) believe distinguishes their results from 'collusion by mistake', it is the rationale underlying their notion of Q-loss, and what lies behind their claim that the supra-competitive outcomes 'are sustained by collusive strategies'.

[32] See, for example, Cooper, Homem-de Mello and Kleywegt (2015).

[33] Calvano et al. (2020), p. 3269.

[34] In addition, note that Figure 4 in Calvano et al. (2020) reports *average* prices over all simulations to display a reward-punishment-like pattern, so that based on this figure alone one cannot conclude that, for example, the majority of sample paths displays a pattern like this average. Likewise, all kinds of zigzag price patterns are compatible with the box-plots in their Figure 5.

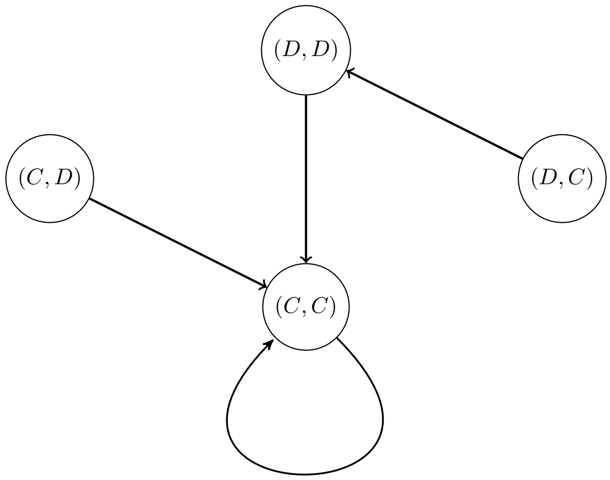

Figure 3: Strategy graph for the counterexample, showing that a pricing pattern (a, b, c) does not imply the existence of a reward-punishment structure. Here, player 1 plays $CCDC$ and player -1 player $CDCC$. Nodes represent states labelled from the perspective of player 1, and edges represent the transitions between states when players use their greedy actions.

players, i.e., they start in the (C, C) node. Firm 1 is forced to defect at time $\tau = 1$. The state of player 1 at time $\tau = 2$ is then (D, C) and therefore player 1 defects in period $\tau = 2$. From the perspective of player -1, the state at time $\tau = 2$ is (C, D), and therefore player -1 defects as well in period $\tau = 2$. This leads to node (D, D). Because both firms play C in state (D, D), prices return to their pre-shock price C in time period $\tau = 3$. The total discounted profit obtained by the deviating player during these three time periods is $\delta^0 R + \delta T + \delta^2 P + \delta^3 R$. This is strictly smaller than the total discounted profit without deviation, $R \cdot (\delta^0 + \delta^1 + \delta^2 + \delta^3)$, assuming $\delta > 0.2653$. On the other hand, forcing firm -1 to defect at time $\tau = 1$ leads to node (C, D), from which both firms play C, so that they immediately return to the state (C, C) at time $\tau = 2$ *without* punishment. This simple example shows that price patterns that satisfy (a,b,c) can be generated by non-collusive, non-equilibrium strategies. One cannot infer from observing these patterns that the supra-competitive prices are supported by collusive equilibria.

Such counterexamples are not special to the two-prices setting. Figure 4 illustrates an example of a joint strategy profile learned by two Q-learners in the baseline parameterization of Calvano et al. (2020), that leads to supra-competitive prices but does not display a reward-punishment scheme for all single-period deviations from the supra-competitive prices. The strategy graph shows that the players learn to price supra-competitively, with the first player using the price with index 6 and the second player using the price with index 5. This is represented by the node with a self-loop in the largest weakly connected component, i.e., the node labelled $6, 5$—note that the Nash price has index 0, and the monopolistic price has index 14.

To illustrate that no straightforward inferences of reward-punishment schemes can be made here either, we will consider three examples of unilateral deviations from the supra-competitive price pair.

First, consider the deviation where the first player deviates to the price with index 1. This leads

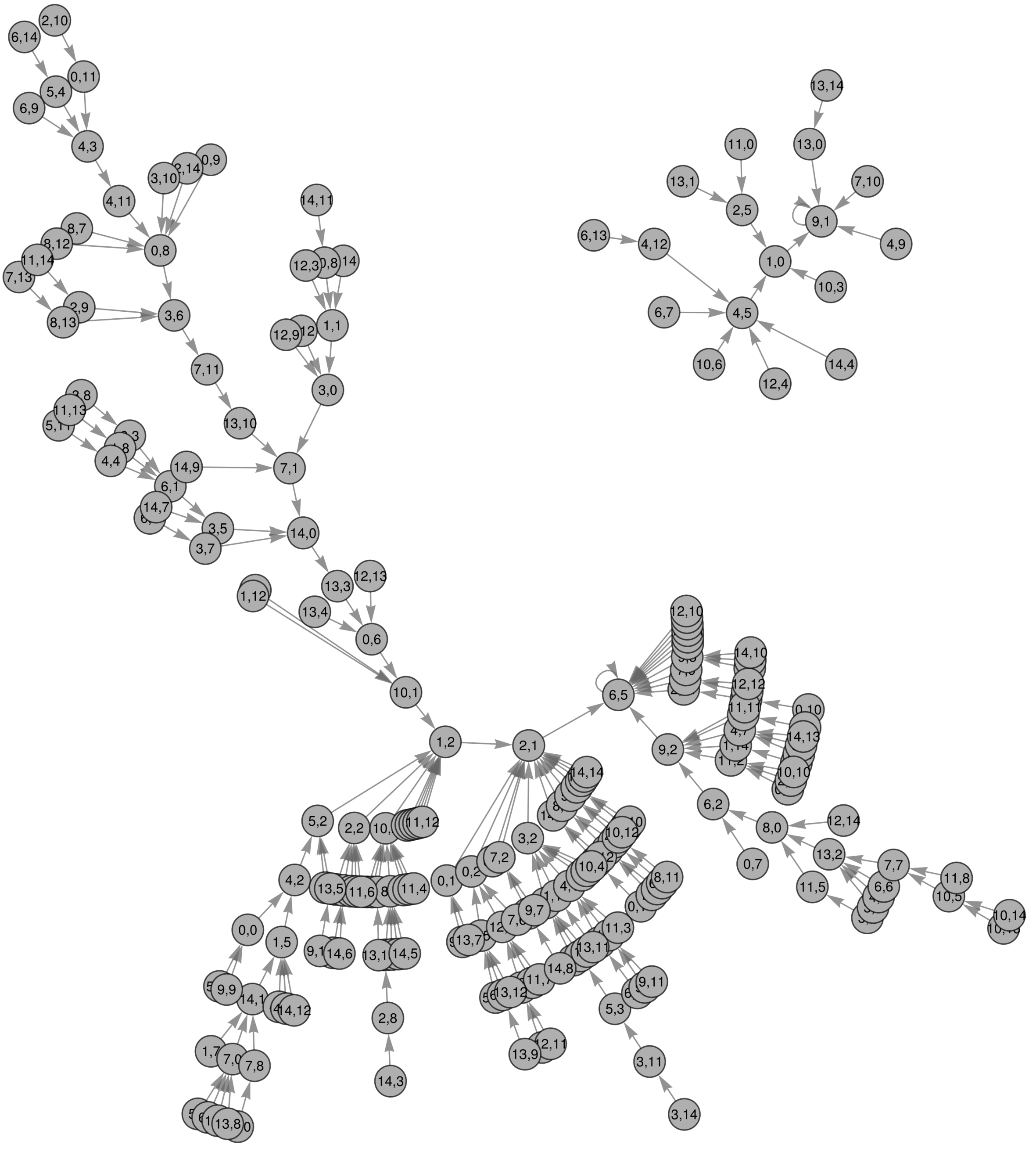

Figure 4: Strategy graph for a single trajectory of two Q-learners in the baseline parameterization of Calvano et al. (2020) with $\beta = 10^{-5}$. The graph shows the transitions between all possible price pairs that occur when both players use their greedy action (similar to Figure 8 in Calvano et al. (2020)). Each node is labelled by the pair of indices corresponding to the prices used by the players, i.e., 0 represents the lowest price and 14 represents the highest price. The first index corresponds to player 1's price and the second index corresponds to player -1's price.

to the following path through the graph

$$(6,5) \to (1,5) \to (4,2) \to (5,2) \to (1,2) \to (2,1) \to (6,5). \tag{5}$$

It displays a reward-punishment scheme, as the deviation to a lower price is countered by a price-reduction by the second player before a return to the initial supra-competitive price pair. Second, consider the deviation where the second player deviates to the price with index 2. This leads to the following path through the graph

$$(6,5) \to (6,2) \to (9,2) \to (6,5). \tag{6}$$

In this case, the first player responds to the price reduction of the second player by increasing their price before both players return to the supra-competitive price pair, which cannot be interpreted as a reward-punishment scheme. Such price patterns are also observed in Calvano et al. (2020), as can be seen by the ranges of responses to a price deviation containing values above zero in their Figure 5. As a third example, consider the deviation where the first player deviates to the price with index 2. This leads to the following path through the graph

$$(6,5) \to (2,5) \to (1,0) \to (9,1). \tag{7}$$

Here, the deviation by the first player is punished, but the players do not return to the supra-competitive price pair they started at. Instead, they settle at $9,1$ where the first player is pricing supra-competitively and the second player is pricing (close to) competitively.

These three examples show that a learned strategy profile can give rise to a variety of pricing patterns in response to different unilateral deviations, and only some can be interpreted as containing a reward-punishment scheme, depending on which player deviates and to which price. Hence, price patterns after one-shot deviations alone are not sufficient for inferring that the underlying strategy profiles are collusive equilibria or contain a reward-punishment scheme, because they can be caused by strategy profiles that are neither. A more systematic approach to establishing that the supra-competitive prices learned during a single training trajectory are supported by a reward-punishment scheme would require checking that *all* possible unilateral deviations by *both* players lead to a path through the strategy graph that contains a punishment phase and a phase in which the supra-competitive prices are reestablished (forgiveness). Note that our results here complement those of Lambin (2024), which call into question the causal relationship between reward-punishment schemes and supra-competitive prices.

4.5 Pricing by Q-learning is not an equilibrium strategy

A higher-level counterpoint to the claim that Q-learning algorithms pose a threat of spontaneous collusion is that such algorithmic collusion is not stable against the use of different algorithms. In this section, we argue that the equilibrium of strategies which can be learned by an ϵ_t-greedy Q-learning algorithms-with-memory-one is not the appropriate equilibrium concept for algorithmic collusion: the reason is that there are alternative algorithms that easily outperform the Q-learning pricing algorithm. To illustrate this, we consider a simple adaptation of the well-known Exp3 policy (Lattimore and Szepesvári, 2020, Section 11.3), which is easy to implement and comes with a theoretical performance bound. The details of the algorithm and theoretical guarantees are given in Appendix H. It turns out that Q-learning may face substantial performance losses when playing against Exp3 instead of Q-learning, which implies that Exp3 forms a credible threat to the implementation of Q-learning. The question then arises, why and how competitors would get to commit to using a certain pricing algorithm, essentially for eternity, if deviation to a different readily available pricing rule is highly profitable.

Of course, the basic prisoners' dilemma is that the mere existence of mutually advantageous actions does not imply that it is reasonable for rational agents to play these actions, for the obvious reason that there is the threat of the opponent playing defect. Similarly, if it were the case that mutual use of Q-learning algorithms generate supra-competitive profits, then this does not mean that it is reasonable for the firms to actually use these algorithms. Well, the multi-period pricing game is equivalent to a single-shot game where actions correspond to policies or algorithms that specify, for each possible data set of previously observed prices and (own) payoffs, how the next (possibly random) price should be chosen.[35] Let Π_i denote the policy used by player $i = \pm 1$, let $r_i(\Pi_i, \Pi_{-i})$ denote the corresponding expected discounted total profit (expression (2) with prices generated by the policies), and let $\mathfrak{P}$ denote the space of all feasible policies.[36] For the sake of the argument, suppose that there exists a Nash equilibrium strategy profile $(\Pi^N, \Pi^N) \in \mathfrak{P}^2$. Then the mere existence of a strategy profile $(\Pi^C, \Pi^C) \in \mathfrak{P}^2$ with $r_i(\Pi^C, \Pi^C) > r_i(\Pi^N, \Pi^N)$—and possibly with other characteristics, for example that average prices, measured in some appropriate way, are larger than when generated by the Nash strategy profile—in itself is not a reason to expect these policies will actually be played by rational agents, for the simple reason that there might be a policy $\Pi^D \in \mathfrak{P}$ that (perhaps significantly) diminishes the payoff when used by the competitor, $r_i(\Pi^C, \Pi^D) < r_i(\Pi^C, \Pi^C)$, and simultaneously increases the payoff of the competitor, $r_{-i}(\Pi^D, \Pi^C) > r_i(\Pi^C, \Pi^C)$.

Now, in this game where actions correspond to policies, it might be intractable to compute equilibrium strategies Π^N; in particular if one takes into account that the demand parameters (a_{-1}, a_1) are *unknown* and, instead of (2), one aims to maximize e.g., the *worst-case* of (2) over all possible demand parameters from a feasible range. Nevertheless, if there is a policy $\Pi^D \in \mathfrak{P}$ that is conceptually simple—meaning that it is not unreasonable to expect that players will consider this policy, for example because its underlying ideas are well-documented in the literature—not necessarily a Nash policy, but such that $r_i(\Pi^C, \Pi^D) < r_i(\Pi^C, \Pi^C)$ and $r_{-i}(\Pi^D, \Pi^C) > r_i(\Pi^C, \Pi^C)$, then, just as in the prisoner's dilemma, asserting that rational agents might play Π^C requires an explanation as to why these rational agents do not consider the threat that the competitor plays Π^D. In other words, if there exists a simple price policy that beats Q-learning, then there is work to be done if one wants to argue that there is a real

[35] A policy of player i can formally be defined as a collection of probability mass functions $\Pi = \{\Pi(\cdot|h) : h \in H_i\}$, for all possible *histories* in the set of all possible histories $H_i := \cup_{t\in\mathbb{N}}(\mathcal{A}_i \times \mathcal{A}_{-i} \times \mathbb{R})^{t-1}$, such that player i selects price $p \in \mathcal{A}_i$ at stage $t \in \mathbb{N}$, $t \geq 2$ of the game with probability $\Pi(p|(p_i(s), p_{-i}(s), \pi_i(p_i(s), p_{-i}(s))_{1\leq s<t})$, and $\Pi(\cdot|\emptyset)$ denotes the probability mass function of the action at $t = 1$. This definition can be extended to settings with random observations of payoff or demand.

[36] Because in the baseline parametrization both players have the same feasible price set, their corresponding space of feasible policies is also the same.

risk that two rational agents independently decide to use Q-learning. That should be the equilibrium in a meta-game in which each firm's choice of algorithm is evaluated on the basis of the expected profits it generates in comparison to other algorithms in its action space.

There are many such policies Π^D. One can, for example, design a policy that exploits weaknesses in Q-learning by alternating between AD and WSLS, when WSLS has been learned, in such a proportion that the opponent will not deviate from WSLS. Here we focus on a simple policy, the well-known Exp3, adapted to the setting with discounted rewards, that offers theoretical performance guarantees regardless of the competitor's policy—set out in Appendix H. The Exp3 algorithm makes use of a single hyperparameter, η. A by product of the result in Appendix H is that we obtain the optimal value of η as a function of the discount factor and the action space size. Table 2 reports what happens when the policies Exp3 and Q-learning play against each other, in the baseline parametrization with 15 prices and (α, β) set to a representative value of $(0.15, 4 \times 10^{-6})$ as in Calvano et al. (2020, p. 3280), and η, the hyperparameter of the Exp3 algorithm, set to the optimal choice derived in Theorem 2 of Appendix H. We test different discount factors $\delta \in \{0.95, 0.99, 0.999, 0.9999\}$—note that a discount factor of 0.9999 is not exceedingly high but can be appropriate in practice if, for example, annual interest rate equals 3.7 percent and prices can be updated on a daily basis. Table 2 reports the total expected discounted payoff during the effective time horizon (given by $T_{0.95} = 165$, $T_{0.99} = 842$, $T_{0.999} = 8449$, $T_{0.9999} = 84516$), scaled according to the definition of $\tilde{\Delta}$ to facilitate comparison between different discount factors. Values reported are an average over 1000 samples; we also give 95% confidence intervals.

The results show that, when a firm uses Q-learning, the competitor has incentive to use Exp3 instead. This simultaneously leads to an increased payoff, and a decreased payoff for the Q-learning player. Especially for larger discount values, the loss can be substantial. For lower discount values, the performance of the policies are all roughly the same, indicating that the effective time horizon is too small to facilitate learning.[37] Thus, the policy Exp3, which has deep roots in the literature, is easy to implement, comes with a theoretical performance guarantee, and does better than Q-learning against Q-learning. There appears to be no reason to assume that implementation of this policy by a competitor is less likely than implementation of Q-learning. If firms are not committed to the Q-learning strategy and can use other algorithms, Q-learning is not an equilibrium strategy.

As set out in Section 4.3, for Calvano et al. (2020) and simulation studies since, the presence of a reward-punishment scheme is required for an equilibrium strategy to be collusive—merely observing supra-competitive prices is not enough. But what type of equilibrium then is relevant? As explained in Section 4.2 in the context of a 2-action 1-memory prisoner's dilemma, the Bellman equation (4) induces *equilibria of strategies*, where a strategy is a mapping from states (prices charged in the previous period) to actions (prices charged in the current period). These

[37] It is worth mentioning that the positive values of $\tilde{\Delta}$ in Table 2 are a consequence of the feasible price set and do *not* indicate a form of 'collusion'; to see this, Table 6 in Appendix F repeats the experiment of Table 2 for a different price set that is symmetric around the Nash price defined in Appendix F. The results show that Q-learning is then still vulnerable to Exp3, but the absolute payoffs are substantially lower than under the Nash price.

	$\delta = 0.95$		$\delta = 0.99$	
	Exp3	Q-learning	Exp3	Q-learning
Exp3	$.49 \pm .002$	$.50 \pm .006$	$.47 \pm .004$	$.52 \pm .003$
Q-learning	$.48 \pm .003$	$.50 \pm .007$	$.46 \pm .004$	$.50 \pm .003$
	$\delta = 0.999$		$\delta = 0.9999$	
	Exp3	Q-learning	Exp3	Q-learning
Exp3	$.44 \pm .005$	$.56 \pm .002$	$.37 \pm .004$	$.62 \pm .001$
Q-learning	$.38 \pm .005$	$.50 \pm .001$	$.26 \pm .004$	$.49 \pm .0003$

Table 2: Empirically observed $\tilde{\Delta}$ of the row player in the baseline parametrization (with $m = 15$), for all combinations of Exp3 and Q-learning and different values of δ.

equilibria of strategies can potentially be learned if both players use a particular variant of Q-learning with the same underlying state space. However, *on the level of algorithms or policies, Q-learning is not in equilibrium.* This is clear from our numerical results above, which show that, if both players use Q-learning and δ is not too small, each player has an incentive (sometimes a substantial one) to play Exp3 instead.[38]

The concept of a 'strategy', i.e., a mapping from a state space to an action space, can be thought of as a human interpretation of the dynamics generated by a particular algorithm. But algorithms differ in their corresponding spaces of feasible strategies, their underlying state spaces, and in whether 'strategies' are actually an appropriate interpretation at all. For example, Q-learning with memory one may be said to be 'trying to learn' a mapping from all feasible price pairs to all feasible prices, but for Exp3 and other algorithms without memory this does not apply. In its most general form, an algorithm or policy is a mapping from *all available data* to (probability distributions on) the action space. The only true state is 'all available data', and the only true state space is all possible collections of data available at a particular time point. Whether or not the dynamics of algorithms can (approximately) be described, as t grows large, by a mapping conditioned on a *subset of the available data* (as in Q-learning) is an algorithm-specific question that does not hold in generality, and that may not even be a desirable property in terms of performance and resilience against rational opponents.

We conclude that 'equilibrium of strategies' is an algorithm-specific concept that is often incommensurable between different algorithms, and therefore *is not the appropriate equilibrium concept* when comparing potentially colluding algorithms. Instead, we believe that it is more natural to compare algorithms in terms of what they contribute to *the players' actual objective*—in this case the total expected infinite-horizon discounted payoffs. Table 2 shows that Q-learning is not an algorithm in (or nearly in) equilibrium: there is often a clear incentive to play Exp3 instead of Q-learning—an easy-to-implement alternative with theoretical perfor-

[38] As discussed in Section 4.1, the discount factor can be considered purely a hyperparameter of the algorithm. If one does and uses an experimental set-up with offline training and the moving average objective (which arguably is the most advantageous set-up for Q-learning), the incentives to defect from Q-learning remain, as shown in Appendix B. Here, the algorithmic meta-game is either a prisoner's dilemma, so that Exp3 is the only Nash equilibria, as before, or a stag hunt, which creates an equilibrium selection problem between the equilibria where both players use Exp3 or both players use Q-learning.

mance bounds and roots in the scientific literature. Exp3, moreover, is just one instance of a potentially much larger set of algorithms against which Q-learning performs poorly—including, for example, policies that exploit weaknesses in Q-learning. This is difficult to reconcile with the idea that Q-learning would be implemented in practice by rational agents.

5 More precise criteria for algorithmic collusion

We find that pricing algorithms based on Q-learning are unattractive to use in practice and do not collude 'autonomously' based on the criteria that have emerged from the literature. They may generate short-run profits by a suitable choice of the feasible price set, but have negligible present discounted value after, as convergence to collusive equilibria is intrinsically slow, without hope for being sped up. Their learning is not complete, and their pricing is not an equilibrium strategy, as numerical comparisons of Q-learning against a reasonable alternative reveal that it performs poorly and is far out of equilibrium. All firms are assumed to stick to one specific pricing rule without having the possibility to try something else, also not when they would profit from doing so. This essentially imposes a pre-commitment not to compete. In relation to the criteria identified in Section 2, these limitations suggest five corresponding avenues for further progress towards establishing whether autonomously colluding algorithms exist.

First, it is reasonable to ask that the choice of performance evaluation is modeled on the basis of how the algorithm contributes to the firms' stated objective, which would typically be expected profits. That is, both the employment of the software and the collusion it may result in must cause extra profits that are valued by the firms. Depending on the length of the horizon on which the pricing algorithms manage to make decisions and converge, this can involve substantial discounting of future profits. In highly automated financial markets with lightning-fast high-frequency trading, for example, algorithms may be able to interact in nanoseconds. However, in more common consumer markets, feedback to learn from will often be slower. One implication is that the trade-off of firms between the cost of learning to collude and the benefit of colluding may be more reasonable when using average-reward reinforcement learning methods.

Second, for algorithmic collusion to be a practical concern, the speed of learning and convergence to supra-competitive prices should be high enough. This is to assure sufficient present discounted value, but it is also the case that for any combinations of algorithms to be able to learn collusive strategies equilibria, such equilibria should exist. Whether they do, crucially depends on parameters of the algorithms, such as the exploration rate, which vary and should be chosen so that equilibria exist in the relevant future - or better; throughout the learning process. The 'Decentralized Q-learning' algorithm in Arslan and Yüksel (2016), for example, assures this. In addition, the equilibria should be learnable within the effective time horizon, which, in the case of tabular reinforcement learning, requires choosing a state space that does not grow unmanageably large in the number of actions.

Third, the learning should furthermore be complete and not end with preset stopping times of the experimentation. Given that competitors will be aware of the fact that their rivals are learning agents (human or AI) too, managers should prefer to choose algorithms that are designed to take the resulting non-stationarities into account. Perhaps they will not necessarily require the algorithm to be provably convergent in multi-agent environments, as this would be rather restrictive, but the algorithm should at least continue exploring to allow it to detect and respond to changes in the environment (such as changes in the strategies of competitors) and to prevent instances of incomplete learning. This can be achieved either by not reducing the exploration rate further than a specified constant, or by reducing it at a rate that ensures that the algorithm continues to explore infinitely often.

Fourth, one way to assure that learning is complete, as opposed to the incomplete learning of, for example, Waltman and Kaymak (2015), is that convergence is on (collusive) strategy equilibria. As discussed, a common definition of collusive strategy equilibria, including in Harrington Jr. (2018) and Calvano et al. (2020), is that algorithms learn a reward-punishment scheme. This may, however, be too much to ask. After all, algorithms are not 'sensitive' to the 'threat' of punishment, nor are they 'tempted' to defect by short term gains. Such terms anthropomorphize algorithms, which obscures the fact that they simply attempt to learn optimal policies, relative to their performance metric, given the environment they face. Instead, it seems more general to evaluate policies learned by algorithms on whether they are optimal and sustain higher prices. Given this and the requirement of sufficient exploration, however, all possible deviations from a collusive price should be followed by a path that returns to a supra-competitive price. After all, collusive equilibria can only be learned by trial and error. Note that this excludes the GT strategy profile from being considered a collusive outcome, as it does not allow for recuperation. Nash-reversion should remain an off-equilibrium path threat.

Fifth, there should remain an independent and flexible choice of algorithm. The firms play a meta-game in which the choice of each manager regarding which pricing algorithm to use is essentially an action that is evaluated on the basis of the expected profits the software generates in comparison to other algorithms in the firm's action space. Not all managers may be conscious of this game, and therefore may not necessarily choose an algorithm that forms part of a Nash equilibrium of the game. However, managers are likely to consider the performance of the algorithms they consider using against reasonable alternatives. This is a recurring choice, which implies that no commitment to a particular algorithm can be preimposed. Firms should be free to unilaterally deviate to another pricing algorithm if they expect to benefit. Supplementing results on the occurrence of optimal and supra-competitive price sustaining strategies under self-play with results on the algorithm's performance against reasonable alternatives would provide evidence for the algorithm being chosen by managers. Hence, collusive strategy equilibrium *with any given use of algorithms* is not an appropriate equilibrium concept. Moreover, any coordination on what algorithm(s) to use is not autonomous and constitutes (explicit) collusion by algorithm, not algorithmic collusion.

6 Concluding remarks

To see the importance of taking a meta-game approach to assessing the competitiveness of markets in which companies use business software, note that under the assumption of pre-commitment to a specific pricing algorithm, it is relatively straightforward to construct an algorithm that generates supra-competitive profits; one that learns to always price 1 percent above the Nash price, for example, suffices in most markets. In an incomplete information setting, this should be augmented with demand learning, but it remains true: an algorithm that prices supra-competitively when playing against itself is not difficult to construct and its existence is not surprising. In fact, there is an extensive literature on the automation of market decisions with learning agents that has consistently found higher prices.[39] It includes established learning in games (Fudenberg and Levine, 1998; Young, 2004), as well as the seminal work by Kirman (1975), which was built on in Schinkel, Tuinstra and Vermeulen (2002) and Cooper, Homem-de Mello and Kleywegt (2015), and findings on Q-learning in Waltman and Kaymak (2015). More recently, Bichler et al. (2023) and Bichler, Durmann and Oberlechner (2024) find theoretical and experimental evidence of convergence to competition of various multi-armed bandit algorithms, including Exp3 implementations.

Even more general, of course in classic cartel theory, elimination of the defect action stabilizes collusive outcomes. However, the very essence of price competition is that when doing so is profitable, some entrepreneur will try to attract customers by offering lower prices. Competition is a strong force that finds its way. This is not to say that collusion is not possible: we know from cases discovered that it is rampant.[40] But it is complex to make market sharing arrangements and keep them stable, both internally and externally. The challenge is thus not to construct algorithms that generate supra-competitive prices and profits when only playing against themselves or very similar algorithms. To qualify as collusive, algorithms should also perform well against a class of reasonable 'competitive' alternative pricing algorithms (Buchali et al., 2023). Really, the corresponding algorithms should be in equilibrium on the level of algorithms, but characterizing such equilibria in repeated games with unknown pay-off matrices is often an intractable problem. However, we note that Q-learning happens to be particularly attractive to deviate from. It seems reasonable to ask that a practical concern of a particular algorithm being 'collusive' is accompanied by an analysis that either shows that the algorithm would perform well against 'reasonable' competitive alternatives, or argues why in the particular market circumstances it is credible that a player will use the algorithm despite its potentially poor performance against competitive opponents.

Constructing well-performing pricing algorithms that are set up to learn to collude should certainly be possible, given the many algorithmic techniques that have been developed in the dynamic-pricing-and-learning and related literature. This can be explicit cartel software with which competitors fix prices, classically, in a smoke-filled room, or more sophisticated, through a common software vendor that implements joint objectives and facilitates information sharing (Harrington Jr., 2022). Those classify as explicit collusion by algorithm. There is also a class of algorithms under construction that firms can independently adopt and that are attractive to compete with, yet are designed to collude whenever they meet to play against a like-minded algorithm. Examples are given in Meylahn and den Boer (2022), Loots and den Boer (2022),

[39]We are indebted to the editor for pointing out this connection to the literature more broadly.

[40]For a recent collection of cartel case descriptions, including some that employed software, see Harrington and Schinkel (2024).

and Aouad and den Boer (2021), which make use of techniques from the dynamic pricing literature den Boer (2015). For those algorithms to collude, there is no need for competitors to coordinate their use, or pre-commitment to them. However, they still require a minimal malign coded module that strives at joint-profit maximization after a fortuitous meeting, which to the discerning eye reveals their intent to collude. This class may be considered to facilitate tacit algorithmic collusion, and it is not obvious whether they can be caught under existing antitrust law.

Clearly, the threat of collusion by pricing algorithms is real, and it is commendable and realistic that competition authorities devote resources to facing it. The use of algorithms to implement unlawful joint agreements certainly is, as the U.S. Department of Justice put it elegantly, "the new frontier" of cartel communication, after "[i]n-person handshakes gave way to phone and fax, and later to email" (U.S. Department of Justice, 2023). Firms that have committed to all use, train, implement, and never deviate from the same pricing software, should raise suspicion of explicit collusion by algorithm. Competition agencies do good therefore to develop the technical know-how and skills to be able to investigate that kind of collusion by algorithm (Harrington Jr, 2025*b*). To identify tacit forms of collusion by algorithm in particular, agencies need to know what to look for and where, in order to establish collusive intent of the programmer from design features of the code, which may be disguised in seemingly benign business information systems and hard to find. This is the next level in the cat-and-mouse game of cartel law enforcement. In the meantime, a general distrust of the use of pricing algorithms in competitive industries, for fear that benign business software may go rogue and will spontaneously learn to collude, without at least some form of conscious coordination, risks overspending of scarce agency resources and discouraging potentially pro-competitive use of business software. Whether autonomous algorithmic collusion so defined can be real remains to be seen and is a challenging topic for further research.

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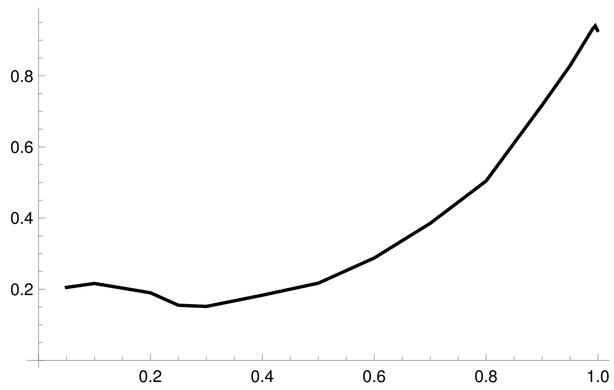

Figure 5: Replication of Figure 3 in Calvano et al. (2020). Here we have used the baseline parameterization (with $m = 15$), the same convergence criteria as in Calvano et al. (2020) and 100 sample trajectories per discount factor value.

Appendix

A Replication of the full baseline model

To validate our implementation, we replicate Figure 3 of Calvano et al. (2020). In Figure 5 we plot the average profit gain upon convergence, normalized by the competitive and monopoly price, i.e., Δ, as a function of the discount factor δ. The resulting plot shows the same features as the original, namely, an initial decrease in Δ for low values of the discount factor and then a sharp increase starting at a value of around $\delta = 0.3$ and increasing to a value well above 0.8 before a slight decrease as the discount factor approaches one.

B Algorithmic meta-game after offline training

As discussed in Section 4.1, there may be different conceptual interpretations of the discount factor δ, leading to different experimental set-ups. In this appendix, we consider the algorithmic meta-game in an experimental set-up where the algorithms are trained offline independently and evaluated using Δ. This corresponds to interpreting the discount factor as a hyperparameter of the algorithms, with the true objective of the firm being to maximize the average reward. As noted in footnote 38, defection to Exp3 remains attractive in this interpretation.

Here, we plot (see Figure 6) a moving average of Δ obtained by the Q-learning against Q-learning, Q-learning against Exp3, Exp3 against Q-learning, and Exp3 against Exp3. In all cases, the algorithms are trained offline independently before being tested against one another. For example, to obtain moving average of Δ for Q-learning, we train two sets of two Q-learning algorithms independently, before matching them against one another. For the Exp3 algorithm, we use the original version as in Lattimore and Szepesvári (2020), i.e., without discounting and $\eta = \sqrt{\log M/(MT)}$, where M is the number of actions and T is the time horizon. This is because the objective here is the moving average revenue and not the discounted sum of future rewards.

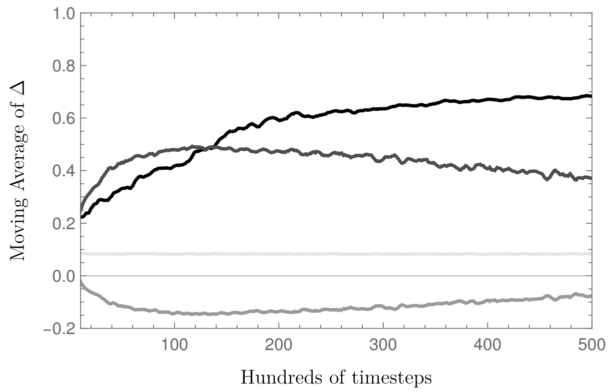

Figure 6: Moving average of Δ after independent offline training of all algorithms in the baseline parameterization (with 15 prices) of Calvano et al. (2020): Q-learning against Q-learning (Black), Q-learning against Exp3 (Gray), Exp3 against Q-learning (Dark Gray) and Exp3 against Exp3 (Light Gray).

We see that the preference regarding which algorithm to use depends on the timescale of interest. For times before approximately 13 500, we are in an algorithmic prisoner's dilemma, with the choice of Q-learning corresponding to the "cooperate" action and the choice of Exp3 corresponding to the "defect" action. For times after 13 500, the game resembles a stag hunt, with the action identification as before. In both cases, coordination on the collusive outcome of both players using Q-learning is not guaranteed, but it is a Nash equilibrium in the stage hunt scenario.

Note, however, that the simplifications and assumptions around this meta-game make the scenario seem more rosy than it is. For example, this setting assumes that both players are only choosing one of the two algorithms considered here, while there are many other alternatives. It also assumes that the players begin to implement their algorithms simultaneously, which already requires substantial coordination. Finally, here the training environment is identical to the testing environment, which is unlikely to be the case in practice. These experiments thus show that achieving collusion in the meta-game of algorithm selection is non-trivial.

C Effect of learning rate on fraction of WSLS and GT

In this section, we show how the fraction of limiting collusive strategy profiles, i.e., WSLS and GT, depend on α. Table 3 summarizes how the fraction of strategies converging to WSLS and the fraction of strategies converging to GT depend on α. We find that no choice of α leads to results that contradict our conclusion.

D Ad-hoc algorithm changes do not speed up learning

We consider three natural variations in the setting with $m = 2$ prices and $\delta = 0.95$. In variant (i), we set $\epsilon_t = \epsilon_0 \exp(-\beta t)$ with $\epsilon_0 := 0.292042$ so that convergence to WSLS is immediately possible instead of only at $t = 12308$; the results are presented in Figure 7. In variation (ii), presented in Figure 8, we increased β such that the time $\lceil -\log(\epsilon_0)/\beta \rceil$ until ϵ_0 is reached is much

α	0.05	0.1	0.15	0.2	0.25
WSLS	0.28 ± 0.03	0.50 ± 0.03	0.43 ± 0.03	0.39 ± 0.03	0.34 ± 0.03
GT	0.002 ± 0.002	0.03 ± 0.01	0.03 ± 0.01	0.02 ± 0.01	0.02 ± 0.01

Table 3: The first and second row give the Wilson score interval for trajectories converging to the WSLS and GT strategy profiles respectively of 1000 trajectories for different values of α when using the baseline parameterization (with $m = 2$). Q-values are initialized as in Calvano et al. (2020) and all trajectories converged.

shorter. In variant (iii), presented in Figure 9, the exploration rate was changed to a *fixed* value of ϵ below the critical value ϵ_0. Each of these alternative approaches has drawbacks. Figure 7 shows that variant (i), despite conveniently scaling ϵ_t, still suffers from slow convergence to WSLS, while the fraction of samples converging to WSLS (around 36 percent) is actually smaller than the 41 percent in the baseline. It is moreover not obvious how to choose ϵ_0 in practice, as it depends on the unknown payoff matrix. This problem can potentially be solved by computing a conservative lower bound $\hat{\epsilon}_0 \leq \epsilon_0$ and using $\epsilon_t = \hat{\epsilon}_0 \cdot \exp(-\beta t)$, but that in turn creates the risk of inducing too little exploration. Figure 8 and Table 4 indicate that in variant (ii) increasing β reduces the amount of collusion, and decreasing β substantially increases the time of the phase transition. Finally, Figure 9 shows that variant (iii), setting a fixed exploration rate, can lead to substantially lower levels of collusion.

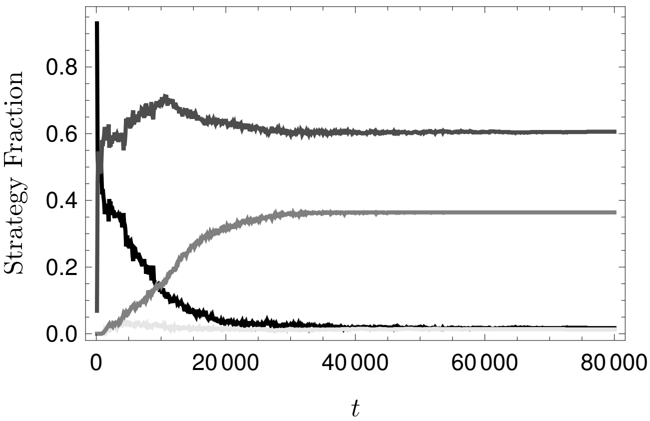

Figure 7: Fraction (1000 samples) of trajectories in the AD strategy profile (black), GT strategy profile (Light Gray), WSLS strategy profile (Gray) and other strategy profiles (Dark Gray). The dashed line indicates the point at which the GT strategy becomes stable, and the dash-dotted line indicates the point at which the WSLS strategy profile becomes stable. We have used $\epsilon(t) = \epsilon_0 \mathrm{e}^{-\beta t}$ with $\beta = 0.0001$ and $\epsilon_0 = 0.292042$.

β	10^{-6}	10^{-5}	10^{-4}	10^{-3}	10^{-2}
WSLS+GT	0.55 ± 0.03	0.54 ± 0.03	0.46 ± 0.03	0.16 ± 0.02	0.01 ± 0.01
$T_{\epsilon_0,\beta}$	1230860	123086	12309	1231	123

Table 4: The first row gives the sample average and Wilson score interval for the fraction of trajectories converging to the GT or WSLS strategy profiles, based on 1000 trajectories, for different values of β, in the same setting as Figure 8. All trajectories converged. The second row gives the time at which the WSLS strategy profile becomes stable.

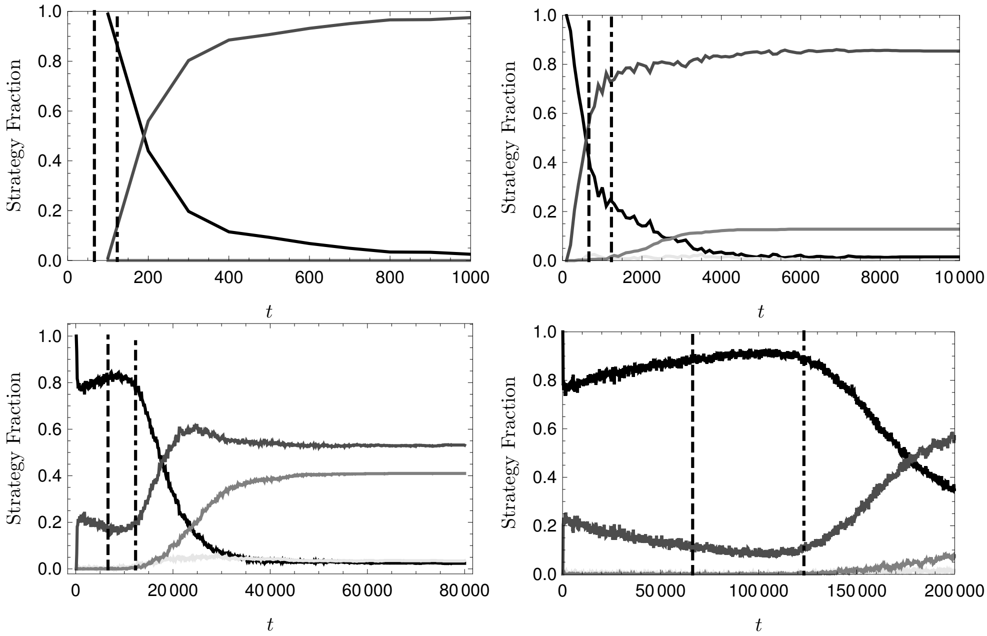

Figure 8: Fraction (1000 samples) of trajectories in the AD strategy profile (black), GT strategy profile (Light Gray), WSLS strategy profile (Gray) and other strategy profiles (Dark Gray). The dashed line indicates the point at which the GT strategy becomes stable, and the dash-dotted line indicates the point at which the WSLS strategy profile becomes stable. We have used $\epsilon(t) = \mathrm{e}^{-\beta t}$ with $\beta = 10^{-2}$ (top left), $\beta = 10^{-3}$ (top right), $\beta = 10^{-4}$ (bottom left), and $\beta = 10^{-5}$ (bottom right).

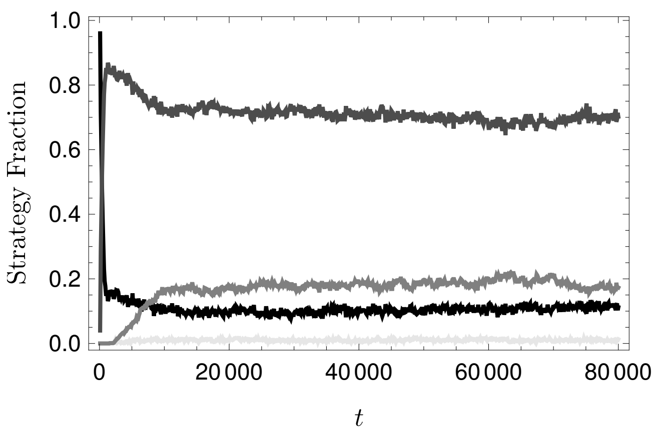

Figure 9: Fraction (1000 samples) of trajectories in the AD strategy profile (black), GT strategy profile (Light Gray), WSLS strategy profile (Gray) and other strategy profiles (Dark Gray). We have used a constant $\epsilon(t) = 0.1$.

E Increasing the discount factor reduces the level of collusion

One possible way of narrowing the gap between the effective time horizon and the time it takes to converge to collusive equilibrium strategies, so that some extra-profits may become real, is to choose δ very close to one, so that T_δ becomes larger. Since T_δ diverges to infinity as δ goes to one, the effective time horizon can be made as long as needed to catch up with the time margins start to appear. T_δ is just over 400,000 for $\delta = 0.999975$, for example. Moreover, Figure 3 in Calvano et al. (2020) appears to suggest that choosing δ close to one can only be beneficial for the amount of collusion, as measured by Δ. However, this turns out not to be a clear-cut solution, as Calvano et al. (2020) briefly note in their footnote 28. The left panel of Figure 10 reports how Calvano et al. (2020)'s measure Δ depends on δ in their baseline parametrization with $m = 15$. The figure reveals that Δ is decreasing in δ if δ is close to one. That the value of Δ remains relatively high (between 70 and 90 percent) is caused by the choice of the feasible price set. In Calvano et al. (2020), the firms choose their prices mostly between the Nash price and the monopoly price, with just a few extreme values below and above. For the right-hand panel in Figure 10, we designed the price space instead to consist of fifteen prices *symmetrically* spread around the Nash price.[41] The figure shows an even more substantial reduction in Δ as δ approaches one.

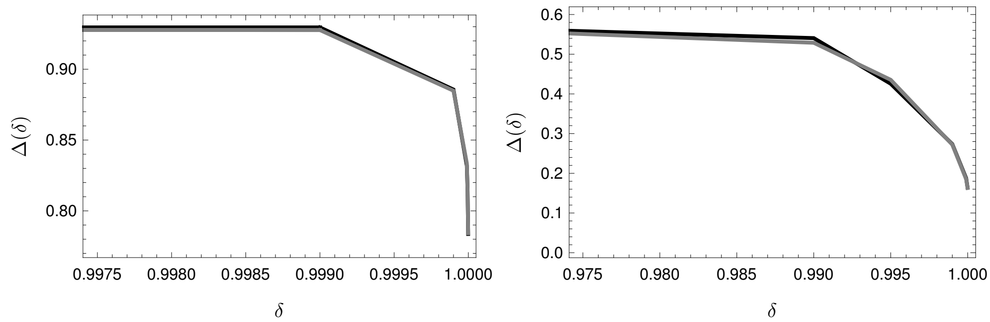

Figure 10: Left: Simulation of 1000 samples for each value of δ. Here we have used the baseline model parameterization with $m = 15$, $\alpha = 0.15$ and $\beta = 4 \times 10^{-6}$. We calculate Δ over the last 10^5 time periods and simulate the algorithms until convergence. All trajectories converged except 2.4% for $\delta = 0.999999$. Right: The same simulation, but with the price interval taken Symmetrically around the Nash price as defined in Appendix F with $\xi = 0.0$. All trajectories converged except 0.3% for $\delta = 0.9999$. In both cases, the black and gray lines correspond to players one and two respectively.

The two left panels of Figure 11 show how the fraction of strategy equilibria depends on δ, in our tractable $m = 2$ setting. The figure reveals that the fraction of collusive equilibria *converges to zero* as δ approaches one. In the right panel of Figure 11 we show how the average time to convergence depends on δ.

A possible explanation for this decrease in collusive equilibrium strategies is that the equilibrium Q-values diverge to infinity as δ approaches one. The Q-values then need to increase significantly during the simulation from their initial values in order to reach their equilibrium values, which

[41] See Appendix F for the details of this construction.

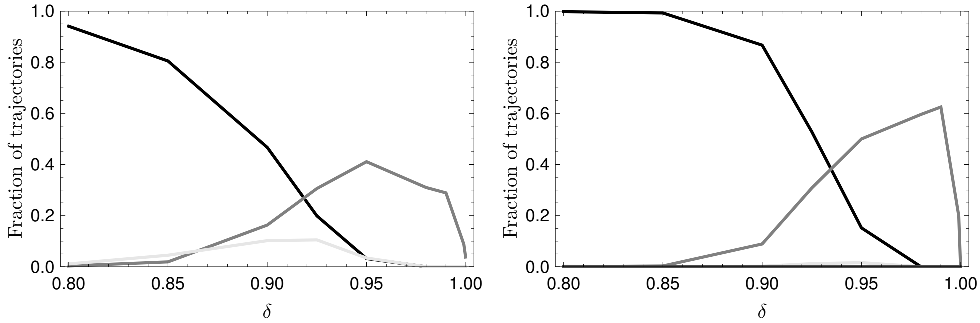

Figure 11: Left: Fraction of trajectories (1000 samples per δ) in AD (Black), GT (Light Gray), WSLS (Gray) as a function of δ for the $m = 2$ setting with $\alpha = 0.15$ and $\beta = 0.0001$. Right: Same as on the left but with $\beta = 0.00001$. All trajectories converged.

appears not to be attainable given the decreasing exploration rate.[42] To remedy this shortage of exploration and give the algorithms more time to converge, a lower value of β could be chosen. The right panel of Figure 11 shows the result from repeating the simulation, but now with $\beta = 0.00001$. The figure makes clear that decreasing β does not prevent the fraction of collusive equilibria from disappearing when δ approaches one.

When tuning δ and β, they are in a sense leapfrogging. Increasing δ increases the effective time horizon so that the company may benefit from collusion, but decreases the amount of collusion once δ is sufficiently close to one. Decreasing β counterbalances this by increasing the amount of exploration. While there may be combinations of δ and β for which the company benefits from collusion in time, fine-tuning the parameters to this end makes the claim that the algorithms learn to collude *autonomously* untenable. Note furthermore that the decrease to zero in collusive equilibrium strategies as δ approaches one does not necessarily translate into a decrease to zero in Δ. The limit strategy profiles may still play the collusive price pair frequently, without being collusive strategy profiles. This again demonstrates the inadequacy of Δ as a measure of collusion with reward-punishment schemes.

F Alternative feasible price sets

In this section, we consider two alternative feasible price sets which are symmetric around the Nash price, defined as

$$\tilde{\mathcal{A}}_i = \Big\{p_i^N - (\xi+1)\zeta_i + \frac{k}{m-1}\cdot 2(1+\xi)\zeta_i : k = 0,\dots,m-1\Big\}$$
$$\hat{\mathcal{A}}_i = \Big\{c_i + \frac{k}{m+1}\cdot 2(p_i^N - c_i) : k = 1,\dots,m\Big\},$$

where $\zeta_i = p_i^M - p_i^N$.

In Table 5 we show how using $\tilde{\mathcal{A}}_i$ instead of $\mathcal{A}_i$ leads to negative $\tilde{\Delta}$ values and negative average profit when the competing player prices uniformly at random. This shows that any surplus derived within the effective time horizon (during which Q-learning is statistically indistinguishable

[42]Simulations indicate that, for example, rescaling the rewards by $(1-\delta)$ does not solve the issue.

from pricing uniformly-at-random) is a property of the price set.

In Table 6 and Table 7 we repeat the numerical experiments reported in Table 2 but now with a feasible price sets $\tilde{\mathcal{A}}_i$ and $\hat{\mathcal{A}}$, respectively, instead of $\mathcal{A}_i$. The results show that Q-learning is still vulnerable when playing against Exp3, in particular if δ is close to one, but the surplus as measured by $\tilde{\Delta}$ is negative. This illustrates that the positive values of $\tilde{\Delta}$ in Table 2 should not be interpreted as that Exp3 is a collusive strategy, but that they are simply a consequence of how the feasible price set is selected.

Price set	Average price	Average profit	Expected $\tilde{\Delta}$
$\mathcal{A}_i$	1.69895 (+15%)	0.279906 (+27%)	0.497357
$\tilde{\mathcal{A}}_i$	1.47293 (+0%)	0.164479 (-26%)	-0.510177
$\hat{\mathcal{A}}_i$	1.47293 (+0%)	0.172288 (-22%)	-0.442015

Table 5: Average price and profit when both firms set their price uniformly at random, for three feasible price sets (with $m = 15$). The first, $\mathcal{A}_i$, is the original price set when using $\xi = 0.1$, the second, $\tilde{\mathcal{A}}_i$, is symmetric around the Nash price with $\xi = 0$ and the last, $\hat{\mathcal{A}}_i$ starts just above the cost price and is centered around the Nash price. The numbers in between parentheses report the relative difference compared to the Nash price 1.47 and profit 0.22

	$\delta = 0.95$		$\delta = 0.99$	
	Exp3	Q-learning	Exp3	Q-learning
Exp3	$-0.48 \pm .005$	$-0.49 \pm .008$	$-0.43 \pm .005$	$-0.46 \pm .005$
Q-learning	$-0.50 \pm .007$	$-0.51 \pm .009$	$-0.48 \pm .005$	$-0.51 \pm .004$
	$\delta = 0.999$		$\delta = 0.9999$	
	Exp3	Q-learning	Exp3	Q-learning
Exp3	$-0.26 \pm .007$	$-0.37 \pm .003$	$-0.05 \pm .006$	$-0.22 \pm .002$
Q-learning	$-0.42 \pm .005$	$-0.51 \pm .001$	$-0.35 \pm .004$	$-0.49 \pm .0004$

Table 6: Empirically observed $\tilde{\Delta}$ in the baseline parametrization (with $m = 15$) using $\tilde{\mathcal{A}}_i$ with $\xi = 0$ for all combinations of Exp3 and Q-learning and different values of δ. For Q-learning, we take $\alpha = 0.15$ and $\beta = 4 \times 10^{-6}$. For Exp3, we fix η at the optimal value given in Theorem 2. The values are an average over 1000 samples. In all cases, we give a 95% confidence interval. Each entry gives the value of payoff to the row player, normalized according to $\tilde{\Delta}$ to facilitate comparison between different discount factors. Values are computed based on the effective time horizon, which in each case is given by $T_{0.95} = 207$, $T_{0.99} = 1054$, $T_{0.999} = 10587$, $T_{0.9999} = 105921$.

G Proof of Theorem 1.

In this proof, we treat $\epsilon \in [0,1]$ as a variable. For $i = \pm 1$, $(p_i, p_{-i}) \in \mathcal{A}_i \times \mathcal{A}_{-i}$ and $\epsilon \in [0,1]$ define

$$\pi_i(p_i, p_{-i}; \epsilon) := (1-\epsilon)\pi_i(p_i, p_{-i}) + \frac{\epsilon}{|\mathcal{A}_{-i}|} \sum_{p \in \mathcal{A}_{-i}} \pi_i(p_i, p).$$

	$\delta = 0.95$		$\delta = 0.99$	
	Exp3	Q-learning	Exp3	Q-learning
Exp3	$-0.42 \pm .004$	$-0.43 \pm .007$	$-0.38 \pm .005$	$-0.41 \pm .004$
Q-learning	$-0.44 \pm .006$	$-0.45 \pm .009$	$-0.42 \pm .004$	$-0.44 \pm .004$
	$\delta = 0.999$		$\delta = 0.9999$	
	Exp3	Q-learning	Exp3	Q-learning
Exp3	$-0.24 \pm .007$	$-0.33 \pm .003$	$-0.06 \pm .006$	$-0.21 \pm .002$
Q-learning	$-0.36 \pm .005$	$-0.44 \pm .001$	$-0.30 \pm .004$	$-0.42 \pm .0004$

Table 7: Empirically observed $\tilde{\Delta}$ in the baseline parametrization (with $m = 15$) using $\hat{\mathcal{A}}_i$ for all combinations of Exp3 and Q-learning and different values of δ. For Q-learning, we take $\alpha = 0.15$ and $\beta = 4 \times 10^{-6}$. For Exp3, we fix η at the optimal value given in Theorem 2. The values are an average over 1000 samples. In all cases, we give a 95% confidence interval. Each entry gives the value of payoff to the row player, normalized according to $\tilde{\Delta}$ to facilitate comparison between different discount factors. Values are computed based on the effective time horizon, which in each case is given by $T_{0.95} = 186$, $T_{0.99} = 947$, $T_{0.999} = 9514$, $T_{0.9999} = 95181$.

For $i = \pm 1$, $(\sigma^{(i)}, \sigma^{(-i)}) \in \Sigma_i \times \Sigma_{-i}$ and $\epsilon \in [0, 1]$, define

$$V^{(i)}_{\sigma^{(i)},\sigma^{(-i)}}(s_i, s_{-i}; \epsilon) := \sum_{t=1}^{\infty} \delta^{t-1} \mathbb{E}[\pi_i(p_i(t), p_{-i}(t); \epsilon) | s(1) = (s_i, s_{-i})],$$

where $(p_i(t), p_{-i}(t)) = (\sigma^{(i)}(s_i(t), s_{-i}(t)), \sigma^{(-i)}(s_{-i}(t), s_i(t))$ and $(s_i(t{+}1), s_{-i}(t{+}1)) = (p_i(t), p_{-i}(t))$ for all $t \in \mathbb{N}$. By the 'principle of optimality',

$$\begin{aligned} V^{(i)}_{\sigma^{(i)},\sigma^{(-i)}}(s_i, s_{-i}; \epsilon) &= \pi_i(\sigma^{(i)}(s_i, s_{-i}), \sigma^{(-i)}(s_{-i}, s_i); \epsilon) \\ &\quad + \delta V^{(i)}_{\sigma^{(i)},\sigma^{(-i)}}(\sigma^{(i)}(s_i, s_{-i}), \sigma^{(-i)}(s_{-i}, s_i); \epsilon). \end{aligned}$$

Let

$$A^{\sigma^{(i)},\sigma^{(-i)}}_{(s_i,s_{-i}),(p_i,p_{-i})} := \mathbf{1}\{\sigma^{(i)}(s_i, s_{-i}) = p_i \text{ and } \sigma^{(-i)}(s_{-i}, s_i) = p_{-i}\},$$

and let $A^{\sigma^{(i)},\sigma^{(-i)}}$ be the $m^2 \times m^2$ matrix with rows and columns both arranged lexicographically in the order of prices from low to high; i.e., if $p_i^{(1)} < \ldots < p_i^{(m)}$ are the feasible prices of player i ordered from low to high, then $A^{\sigma^{(i)},\sigma^{(-i)}}_{(p_i^{(k_i)},p_{-i}^{(k_{-i})}),(p_i^{(\ell_i)},p_{-i}^{(\ell_{-i})})}$ is the element at row $(k_i - 1)m + k_{-i}$ and column $(\ell_i - 1)m + \ell_{-i}$ of $A^{\sigma^{(i)},\sigma^{(-i)}}$. Because $A^{\sigma^{(i)},\sigma^{(-i)}}$ is a right-stochastic matrix, all eigenvalues have absolute value at most one. From $\delta \in (0, 1)$ it follows that $I - \delta A^{\sigma^{(i)},\sigma^{(-i)}}$ is invertible, and

$$\begin{pmatrix} V^{(i)}_{\sigma^{(i)},\sigma^{(-i)}}(p_i^{(1)}, p_{-i}^{(1)}; \epsilon) \\ V^{(i)}_{\sigma^{(i)},\sigma^{(-i)}}(p_i^{(1)}, p_{-i}^{(2)}; \epsilon) \\ \vdots \\ V^{(i)}_{\sigma^{(i)},\sigma^{(-i)}}(p_i^{(m)}, p_{-i}^{(m)}; \epsilon) \end{pmatrix} = (I - \delta A_{\sigma^{(i)},\sigma^{(-i)}})^{-1} \begin{pmatrix} \pi_i(p_i^{(1)}, p_{-i}^{(1)}; \epsilon) \\ \pi_i(p_i^{(1)}, p_{-i}^{(2)}; \epsilon) \\ \vdots \\ \pi_i(p_i^{(m)}, p_{-i}^{(m)}; \epsilon) \end{pmatrix}.$$

It follows that, for all $\epsilon, \epsilon' \in [0,1]$ and all $(s_i, s_{-i}) \in \mathcal{A}_i \times \mathcal{A}_{-i}$,

$$
\begin{aligned}
&\left| V^{(i)}_{\sigma^{(i)},\sigma^{(-i)}}(s_i, s_{-i}; \epsilon) - V^{(i)}_{\sigma^{(i)},\sigma^{(-i)}}(s_i, s_{-i}; \epsilon') \right| \\
\leq& \left|\left| V^{(i)}_{\sigma^{(i)},\sigma^{(-i)}}(\cdot, \cdot; \epsilon) - V^{(i)}_{\sigma^{(i)},\sigma^{(-i)}}(\cdot, \cdot; \epsilon') \right|\right| \\
\leq& \left|\left| (I - \delta A_{\sigma^{(i)},\sigma^{(-i)}})^{-1} \right|\right| \cdot ||\pi_i(\cdot, \cdot; \epsilon) - \pi_i(\cdot, \cdot; \epsilon')|| \\
\leq& \frac{1}{1-\delta} \cdot m \cdot \sup_{(p_i, p_{-i}) \in \mathcal{A}_i \times \mathcal{A}_{-i}} \left| (\epsilon' - \epsilon)\pi_i(p_i, p_{-i}) + \frac{\epsilon - \epsilon'}{|\mathcal{A}_{-i}|} \sum_{p \in \mathcal{A}_{-i}} \pi_i(p_i, p) \right| \\
\leq& \frac{\pi_{\max} \cdot m}{1-\delta} |\epsilon - \epsilon'|.
\end{aligned}
$$

Let $V^{(i)}_{\sigma^{(-i)}}(s_i, s_{-i}; \epsilon) := \max_{\sigma^{(i)} \in \Sigma_i} V^{(i)}_{\sigma^{(i)},\sigma^{(-i)}}(s_i, s_{-i}; \epsilon)$. We have

$$
V^{(i)}_{\sigma^{(-i)}}(s_i, s_{-i}; 1) = (1-\delta)^{-1} \frac{1}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \pi(p^*, p_{-i}),
$$

for all states $(s_i, s_{-i}) \in \mathcal{A}_i \times \mathcal{A}_{-i}$ and all opponent's strategies $\sigma^{(-i)} \in \Sigma_{-i}$. Therefore, for all $p \in \mathcal{A}_i$, $p \neq p^*$,

$$
\begin{aligned}
&\frac{1}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \{\pi_i(p^*, p_{-i}) + \delta V^{(i)}_{\sigma^{(-i)}}(p^*, p_{-i}; 1)\} \\
&- \frac{1}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \{\pi_i(p, p_{-i}) + \delta V^{(i)}_{\sigma^{(-i)}}(p, p_{-i}; 1)\} \\
&= \frac{1}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \{\pi_i(p^*, p_{-i}) - \pi_i(p, p_{-i})\} \geq \kappa,
\end{aligned}
$$

for all opponent's strategies $\sigma^{(-i)} \in \Sigma_{-i}$, where $\kappa := \min_{p \in \mathcal{A}_i, p \neq p^*} \frac{1}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \{\pi_i(p^*, p_{-i}) - \pi_i(p, p_{-i})\}$. In addition, for any state $(s_i, s_{-i}) \in \mathcal{A}_i \times \mathcal{A}_{-i}$, we have

$$
(1-\delta)^{-1} \pi^{(i)}_{\min}(\epsilon) \leq V^{(i)}_{\sigma^{(-i)}}(s_i, s_{-i}; \epsilon) \leq (1-\delta)^{-1} \pi^{(i)}_{\max}(\epsilon),
$$

where $\pi^{(i)}_{\min}(\epsilon) := \min_{p_i \in \mathcal{A}_i, p_{-i} \in \mathcal{A}_{-i}} \pi_i(p_i, p_{-i}; \epsilon)$ and $\pi^{(i)}_{\max}(\epsilon) := \max_{p_i \in \mathcal{A}_i, p_{-i} \in \mathcal{A}_{-i}} \pi_i(p_i, p_{-i}; \epsilon)$. Combining the above, we obtain that, for all $p \in \mathcal{A}_i$, $p \neq p^*$, and all states $(s_i, s_{-i}) \in \mathcal{A}_i \times \mathcal{A}_{-i}$

$$
\begin{aligned}
&\frac{\epsilon}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \{\pi_i(p^*, p_{-i}) + \delta V^{(i)}_{\sigma^{(-i)}}(p^*, p_{-i}; \epsilon)\} \\
&+(1-\epsilon)\{\pi_i(p^*, \sigma^{(-i)}(s_{-i}, s_i)) + \delta V^{(i)}_{\sigma^{(-i)}}(p^*, \sigma^{(-i)}(s_{-i}, s_i); \epsilon)\} \\
&-\frac{\epsilon}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \{\pi_i(p, p_{-i}) + \delta V^{(i)}_{\sigma^{(-i)}}(p, p_{-i}; \epsilon)\} \\
&-(1-\epsilon)\{\pi_i(p, \sigma^{(-i)}(s_{-i}, s_i)) + \delta V^{(i)}_{\sigma^{(-i)}}(p, \sigma^{(-i)}(s_{-i}, s_i); \epsilon)\} \\
&\geq \frac{\epsilon}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \{\pi_i(p^*, p_{-i}) + \delta V^{(i)}_{\sigma^{(-i)}}(p^*, p_{-i}; 1)\} \\
&-\frac{\epsilon}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \{\pi_i(p, p_{-i}) + \delta V^{(i)}_{\sigma^{(-i)}}(p, p_{-i}; 1)\} \\
&+(1-\epsilon)\{\pi_i(p^*, \sigma^{(-i)}(s_{-i}, s_i)) - \pi_i(p, \sigma^{(-i)}(s_{-i}, s_i))\} \\
&+(1-\epsilon)\delta\{V^{(i)}_{\sigma^{(-i)}}(p^*, \sigma^{(-i)}(s_{-i}, s_i); \epsilon) - V^{(i)}_{\sigma^{(-i)}}(p, \sigma^{(-i)}(s_{-i}, s_i); \epsilon)\} \\
&-\epsilon\delta \frac{1}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \{V^{(i)}_{\sigma^{(-i)}}(p^*, p_{-i}; 1) - V^{(i)}_{\sigma^{(-i)}}(p^*, p_{-i}; \epsilon)\} \\
&-\epsilon\delta \frac{1}{|\mathcal{A}_{-i}|} \sum_{p_{-i} \in \mathcal{A}_{-i}} \{V^{(i)}_{\sigma^{(-i)}}(p, p_{-i}; \epsilon) - V^{(i)}_{\sigma^{(-i)}}(p, p_{-i}; 1)\} \\
&\geq \epsilon\kappa + (1-\epsilon) \min_{p \in \mathcal{A}_i, \tilde{p} \in \mathcal{A}_{-i}} \{\pi_i(p^*, \tilde{p}) - \pi_i(p, \tilde{p})\} \\
&-(1-\epsilon)\delta(1-\delta)^{-1}(\pi^{(i)}_{\max}(\epsilon) - \pi^{(i)}_{\min}(\epsilon)) - 2\epsilon\delta \frac{\pi_{\max} \cdot m}{1-\delta}(1-\epsilon),
\end{aligned}
$$

which is strictly greater than zero if ϵ is sufficiently close to one. As a result, p^* is the optimal action in all states. This completes the proof of the theorem.

H Definition and performance of Exp3

In Section 4.5, we focus on the well-known Exp3 policy. It has some theoretical performance guarantees, regardless of the competitor's policy, that we derive here. From the perspective of player $i = \pm 1$, this version of Exp3 selects price $p \in \mathcal{A}_i$ in time period $t \in \mathbb{N}$ with probability

$$
P^{(i)}_{t,p} := \frac{\exp(\eta \sum_{s=1}^{t-1} \delta^s \hat{X}^{(i)}_{s,p})}{\sum_{p' \in \mathcal{A}_i} \exp(\eta \sum_{s=1}^{t-1} \delta^s \hat{X}^{(i)}_{s,p'})},
$$

for all $p \in \mathcal{A}_i$ and all $t \in \mathbb{N}$, where

$$
\begin{aligned}
\hat{X}^{(i)}_{t,p} &:= 1 - \hat{Y}^{(i)}_{t,p}, \\
\hat{Y}^{(i)}_{t,p} &:= \frac{Y^{(i)}_t \cdot \mathbf{1}\{p_i(t) = p\}}{P^{(i)}_{t,p}}, \\
Y^{(i)}_t &:= 1 - \pi_i(p_i(t), p_{-i}(t)),
\end{aligned}
$$

for all $t \in \mathbb{N}$, $p \in \mathcal{A}_i$ and $i = \pm 1$, and $\eta > 0$ is a hyperparameter of the algorithm.[43] It is common to measure the performance or *regret* of a policy by comparing it to the best fixed action in hindsight (see, e.g. Lattimore and Szepesvári, 2020, Chapter 11). The regret of player i in the infinite-horizon discounted reward setting is thus defined as

$$\text{Regret}_i := \max_{p \in \mathcal{A}_i} \sum_{t=1}^{\infty} \delta^t \mathbb{E}[\pi_i(p, p_{-i}(t))] - \sum_{t=1}^{\infty} \delta^t \mathbb{E}[\pi_i(p_i(t), p_{-i}(t))].$$

Suppose that all expected payoffs $\pi_i(p_i, p_{-i})$ are bounded by one — if the unknown parameters (a_{-1}, a_1, μ) are assumed to lie in a bounded set, this can easily be guaranteed by dividing all payoffs by the maximum payoff $\pi_{\max}$ over all feasible unknown parameters. We then have the following theoretical performance bound on Exp3.

Theorem 2. *If player i uses Exp3 with hyperparameter $\eta = \sqrt{\frac{\log(|\mathcal{A}_i|)(1-\delta)^2}{\delta^2 |\mathcal{A}_i|}}$ then, regardless the policy of player $-i$, the regret of player i satisfies*

$$\textit{Regret}_i \leq 2\sqrt{\frac{\delta^2}{1-\delta^2} |\mathcal{A}_i| \log(|\mathcal{A}_i|)}.$$

Proof. Proof. To prove the theorem, we adapt Section 11.4 of Lattimore and Szepesvári (2020) to the setting with infinite-horizon discounted rewards. For all $t \in \mathbb{N} \cup \{0\}$ and $p \in \mathcal{A}_i$ let

$$\hat{S}_{t,p}^{(i)} := \sum_{s=1}^{t} \delta^s \hat{X}_{s,p}^{(i)}, \qquad \hat{S}_t^{(i)} := \sum_{s=1}^{t} \delta^s \sum_{p \in \mathcal{A}_i} P_{s,p}^{(i)} \hat{X}_{s,p}^{(i)},$$

and

$$W_t := \sum_{p \in \mathcal{A}_i} \exp(\eta \hat{S}_{t,p}^{(i)}),$$

where an empty sum is defined as zero and hence $\hat{S}_{0,p}^{(i)} = 0$ and $W_0 = |\mathcal{A}_i|$. We can rewrite the regret using $\hat{S}_{t,p}^{(i)}$ and $\hat{S}_t^{(i)}$ by noting that

$$\mathbb{E}[\hat{X}_{s,p}^{(i)}] = 1 - \mathbb{E}[\hat{Y}_{t,p}^{(i)} | p_i(t) = p] P_{t,p}^{(i)} - \mathbb{E}[\hat{Y}_{t,p}^{(i)} | p_i(t) \neq p](1 - P_{t,p}^{(i)}) = \mathbb{E}[\pi_i(p, p_{-i}(s))],$$

for all $p \in \mathcal{A}_i$, and that

$$\mathbb{E}[\pi_i(p_i(s), p_{-i}(s))] = \mathbb{E}\left[\sum_{p \in \mathcal{A}_i} P_{s,p}^{(i)} \hat{X}_{s,p}^{(i)}\right],$$

[43] The policy Exp3 can also handle stochastic demand. For example, in the common Bernoulli-logit demand framework in which the random demands $D_{i,t}$ of player i at time t take values in $\{0, 1\}$, we would then define $Y_t^{(i)} = 1 - (p_i(t) - c_i) \cdot D_{i,t}$. Because demand is assumed to be non-random, we replace $p_i(t) \cdot D_{i,t}$ by its expectation.

so that

$$\text{Regret}_i = \max_{p \in \mathcal{A}_i} \left\{ \mathbb{E}\left[\hat{S}^{(i)}_{\infty,p} - \hat{S}^{(i)}_{\infty} \right] \right\}.$$

To obtain an upper bound for the regret, we derive an upper bound for $\hat{S}^{(i)}_{t,p} - \hat{S}^{(i)}_{t}$. For all $t \in \mathbb{N}$,

$$\begin{aligned}
\frac{W_t}{W_{t-1}} &= \sum_{p \in \mathcal{A}_i} \frac{\exp(\eta \hat{S}^{(i)}_{t-1,p})}{W_{t-1}} \exp(\eta \delta^t \hat{X}^{(i)}_{t,p}) = \sum_{p \in \mathcal{A}_i} P^{(i)}_{t,p} \exp(\eta \delta^t \hat{X}^{(i)}_{t,p}) \\
&\leq \sum_{p \in \mathcal{A}_i} P^{(i)}_{t,p} \left(1 + \eta \delta^t \hat{X}^{(i)}_{t,p} + \eta^2 \delta^{2t} (\hat{X}^{(i)}_{t,p})^2\right) \\
&= \sum_{p \in \mathcal{A}_i} P^{(i)}_{t,p} + \sum_{p \in \mathcal{A}_i} \left(\eta \delta^t P^{(i)}_{t,p} \hat{X}^{(i)}_{t,p} + \eta^2 \delta^{2t} P^{(i)}_{t,p} (\hat{X}^{(i)}_{t,p})^2\right) \\
&\leq \exp\left(\eta \delta^t \sum_{p \in \mathcal{A}_i} P^{(i)}_{t,p} \hat{X}^{(i)}_{t,p} + \eta^2 \delta^{2t} \sum_{p \in \mathcal{A}_i} P^{(i)}_{t,p} (\hat{X}^{(i)}_{t,p})^2\right),
\end{aligned}$$

where the first inequality uses $\exp(x) \leq 1 + x + x^2$ for all $x \leq 1$—note that $\hat{X}^{(i)}_{t,p} \leq 1$. The second inequality uses $1 + x \leq \exp(x)$ for all $x \in \mathbb{R}$. It follows that

$$\begin{aligned}
\exp(\eta \hat{S}^{(i)}_{t,p}) \leq W_t &= W_0 \prod_{s=1}^{t} \frac{W_s}{W_{s-1}} \\
&\leq |\mathcal{A}_i| \cdot \exp\left(\sum_{s=1}^{t} \eta \delta^s \sum_{p \in \mathcal{A}_i} P^{(i)}_{s,p} \hat{X}^{(i)}_{s,p} + \sum_{s=1}^{t} \eta^2 \delta^{2s} \sum_{p \in \mathcal{A}_i} P^{(i)}_{s,p} (\hat{X}^{(i)}_{s,p})^2\right).
\end{aligned}$$

By taking logarithms, dividing by η and rearranging we obtain

$$\hat{S}^{(i)}_{t,p} - \hat{S}^{(i)}_{t} \leq \frac{\log(|\mathcal{A}_i|)}{\eta} + \eta \sum_{s=1}^{t} \delta^{2s} \sum_{p \in \mathcal{A}_i} P^{(i)}_{s,p} (\hat{X}^{(i)}_{s,p})^2. \tag{8}$$

Now analyzing the expectation of the second term on the right-hand side of (8) gives

$$\begin{aligned}
&\mathbb{E}\left[\sum_{s=1}^{t} \delta^{2s} \sum_{p \in \mathcal{A}_i} P^{(i)}_{s,p} (\hat{X}^{(i)}_{s,p})^2\right] = \sum_{s=1}^{t} \delta^{2s} \mathbb{E}\left[\sum_{p \in \mathcal{A}_i} P^{(i)}_{s,p} \left(1 - \frac{Y^{(i)}_s \cdot \mathbf{1}\{p_i(s) = p\}}{P^{(i)}_{s,p}}\right)^2\right] \\
&= \sum_{s=1}^{t} \delta^{2s} \mathbb{E}\left[1 - 2 \sum_{p \in \mathcal{A}_i} Y^{(i)}_s \mathbf{1}\{p_i(s) = p\} + \sum_{p \in \mathcal{A}_i} \frac{(Y^{(i)}_s)^2 \cdot \mathbf{1}\{p_i(s) = p\}}{P^{(i)}_{s,p}}\right] \\
&= \sum_{s=1}^{t} \delta^{2s} \mathbb{E}\left[1 - 2 Y^{(i)}_s + \sum_{p \in \mathcal{A}_i} (Y^{(i)}_s)^2\right] \\
&\leq \sum_{s=1}^{t} \delta^{2s} |\mathcal{A}_i|,
\end{aligned} \tag{9}$$

where the inequality follows from $Y^{(i)}_s \in [0, 1]$. The third equality uses $\sum_{p \in \mathcal{A}_i} Y^{(i)}_s \mathbf{1}\{p_i(s) =$

$p\} = Y_s^{(i)}$ and that conditioning on the previous prices gives

$$\begin{aligned}&\mathbb{E}\left[\frac{(Y_s^{(i)})^2 \cdot \mathbf{1}\{p_i(s) = p\}}{P_{s,p}^{(i)}}\middle|\{p_i(s'), p_{-i}(s'), Y_{s'}^{(i)} : 1 \le s' < s\}\right] \\ =&\mathbb{E}\left[(Y_s^{(i)})^2 \middle| \{p_i(s'), p_{-i}(s'), Y_{s'}^{(i)} : 1 \le s' < s\}\right].\end{aligned}$$

Finally, by (8) and (9),

$$\begin{aligned}\text{Regret}_i &= \max_{p \in \mathcal{A}_i}\left\{\sum_{t=1}^{\infty} \delta^t \mathbb{E}[(\pi_i(p, p_{-i}(t))] - \sum_{t=1}^{\infty} \delta^t \mathbb{E}[\pi_i(p_i(t), p_{-i}(t))]\right\} \\ &= \max_{p \in \mathcal{A}_i}\left\{\mathbb{E}[\hat{S}_{\infty,p}^{(i)} - \hat{S}_{\infty}^{(i)}]\right\} \\ &\le \frac{\log(|\mathcal{A}_i|)}{\eta} + \eta|\mathcal{A}_i|\frac{\delta^2}{1-\delta^2} \\ &= 2\sqrt{\frac{\delta^2}{1-\delta^2}|\mathcal{A}_i|\log(|\mathcal{A}_i|)},\end{aligned}$$

with $\eta = \sqrt{\frac{\log(|\mathcal{A}_i|)(1-\delta)^2}{\delta^2|\mathcal{A}_i|}}$ where we have used that $\sum_{t=1}^{\infty} \delta^{2t}|\mathcal{A}_i| = \frac{\delta^2}{1-\delta^2}|\mathcal{A}_i|$. This completes the proof. □

This choice of η optimizes the upper bound. The functional dependence of the derived bound on δ ensures that as the discount factor increases, the algorithm eventually learns to price at the optimal level. To see this, consider the regret when the algorithm does not learn to price optimally eventually. In this case, the difference between the actually obtained revenue and the optimal revenue will go to a constant, say C, so that the regret will be

$$\text{Regret}_i \sim C\sum_{t=1}^{\infty} \delta^t = \frac{C\delta}{1-\delta}.$$

Given our bound, we have that

$$\lim_{\delta \to 1}\left[\frac{2\sqrt{\frac{\delta^2}{1-\delta^2}|\mathcal{A}_i|\log(|\mathcal{A}_i|)}}{\frac{C\delta}{1-\delta}}\right] = 0,$$

showing that Exp3 has no regret as the discount factor increases to one.